A Non-Generative Account of Consciousness:

Temporal Non-Closure and Cross-Scale Log Referencing

Yuhki Endoh

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Abstract

This paper formulates consciousness not as an internally generated object, but as a state that opens temporarily between processes operating at different time scales. The central claim is that consciousness, in this descriptive layer, is not treated as a specific brain region, representation, or output, but as a log-referenceable relation established between fast predictive processes and slow sedimentary memory processes.

To describe this relation, we introduce internal delay or buffering as τ, an effective torque-like quantity as $\mathfrak{T}$, and information potential as Φ. Speed difference is written $\Delta v = v_f - v_s$ and coupling viscosity as μ. Libet-type experiments, split-brain experiments, qualia, measurement, and non-localization are all reinterpreted from the perspective of inter-scale transformation and incomplete closure.

The paper further redescribes log referencing as non-closed information flow along a meaning gradient. The IFGT variables I, J_I, and Φ are used here through a projected cognitive-window flow $J_{\log}$, not as a direct identification of consciousness with a general information field.

Contents

1 Introduction

Discussions of consciousness typically begin with questions such as: where is it generated, which circuits are responsible, and which representational format does it correspond to? This paper changes the framing of these questions. Consciousness is not treated here as something produced somewhere; it is modeled as a state that opens when processes at different time scales become mutually referenceable while maintaining a certain delay and coupling.

In this view, consciousness is not identical to the self, to thought, or to judgment. It is the condition under which fast predictive processes become referenceable to the layers of slow memory, reporting, and sedimentation. Accordingly, the aim of this paper is not to provide a final definition of consciousness, but to establish a structural position from which consciousness can be rigorously discussed.

Previous supplementary treatments addressed qualia, Libet experiments, split-brain phenomena, measurement, and the localization problem individually. This paper reorganizes them into a single independent theoretical draft.

1.1 The Position of This Paper: Three Declarations

This section states in advance the position this paper takes with respect to existing theories of consciousness, restricted to three points. Detailed comparison is given in §18. The purpose here is to fix the coordinate frame that the reader should maintain throughout the paper.

First: an opening model, not a generative model

This paper does not treat consciousness as an output generated somewhere. Many models — including Global Workspace, IIT, and Higher-Order theories — describe consciousness as the product of a process or as the endpoint of an integration measure. This paper does not adopt that premise as its starting point.

More precisely, consciousness is not modeled here as a generated object. It is treated as a state (a log-referencing state) that opens when a specific speed difference, coupling viscosity, internal delay, and meaning gradient are established between fast predictive processes and slow sedimentary memory processes. This difference is not merely definitional; it concerns the structure of the question itself — what should be asked about consciousness.

Second: description as a temporal regime, not reduction to a single quantity

This paper does not reduce consciousness to a single physical quantity, information measure, or integration index. Consciousness is not reducible to any one of: speed difference Δv, coupling viscosity μ, internal delay τ, effective torque-like quantity $\mathcal{T}_\Phi$, or meaning gradient $\nabla\Phi$. Consciousness is a temporal non-closure regime that obtains when these variables enter a certain relation.

This position is not eliminativist. It does not aim to dissolve consciousness into a description of physical quantities or neural activity. Nor is it dualist. It does not add consciousness as a special entity. It describes consciousness as the dynamic relation among multiple processing layers.

Third: redescription of conditions, not a negation of neuroscience

This paper holds that consciousness does not localize to a single site. This is not a negation of neuroscience. It is entirely possible that specific brain regions, circuits, oscillations, or activity patterns correlate with conscious states. In the framework of this paper, these are interpreted as local conditions that participate in the log-referencing loop.

Neural correlates of consciousness are not consciousness itself. They are part of the local substrate through which consciousness opens. The shift from "finding the seat of consciousness" to "investigating the conditions under which consciousness opens" is the stance this paper takes toward neuroscience.

The reader is invited to hold these three points throughout the discussion that follows. What this paper claims is not the negation or mystification of consciousness, but the fixing of a structural position from which consciousness can be questioned.

Note also that the two-layer structure of fast and slow layers treated in this paper stands on IFGT (Information Field Geometry Theory), Bio-IFGT, and the cognitive-window projection developed in Conceptual Topography. IFGT supplies the general language of information density, flow, and potential; Bio-IFGT explains why such layered dynamics can appear in biological systems; CT describes how conceptual terrains stabilize within a cognitive observation window. The grounds for why this two-layer structure appears in biological systems are given in §4. Even for readers approaching this paper as a standalone work, knowing this connection deepens the meaning of the equations.

2 Intuitive Picture

Intuitively, there are at least two speeds within the system. One is a fast stream handling prediction, reaction, and momentary updating. The other is a slow stream handling memory, stabilization, and long-term structural change.

Consciousness is not the fast stream itself. Nor is it the slow memory layer itself. Consciousness opens at the boundary just before the fast stream fully sediments into the slow layer — or where that sedimentation becomes possible.

This boundary is not a static location. It is a dynamic relation that obtains only when the speed difference between the fast and slow processes is maintained while their coupling does not break. When the speed difference vanishes, the boundary disappears. When the speed difference is too large, referencing ruptures.

3 Notation and Minimal Assumptions

The symbols used in this paper constitute a local working notation; they do not carry over the standard meanings of existing theories.

- t: physical time, externally measurable.
- τ: internal time delay, buffering, retention. Not subjective time itself.
- v_f: fast, predictive and reactive dynamics.

- v_s: slow, sedimentary and stabilizing dynamics.
- $\Delta v = v_f - v_s$: speed difference.
- μ: coupling viscosity between speed scales.
- $\mathfrak{T}$: effective torque-like quantity that transforms speed difference.
- Φ: information potential.
- I: information density.
- $J_{\log}$: log-referencing flow in the consciousness window.

Crucially, τ is not used as torque. τ denotes internal delay; the torque-like quantity is consistently written $\mathfrak{T}$. Both quantities involve $\Delta v/\mu$ in their structural expressions, but they describe different aspects: τ is the temporal thickness of referencing, while $\mathfrak{T}$ is the transformation load on speed difference.

The structural relations among these quantities are organized in §7.

4 Minimal Structure: SSO

4.1 Why Two Layers Emerge: Grounding in Bio-IFGT

Before entering this section, one point must be fixed.

The two-layer structure of fast and slow layers is not a distinction introduced for the convenience of explaining consciousness. This structure emerges necessarily from the process by which life attractors form.

Bio-IFGT describes life as a "dynamic quasi-closure structure created by information flow." For life to maintain a difference from its environment while sustaining itself, at least two processing operations at different time scales are required.

The first is fast processing, which handles immediate environmental response. It is involved in prediction, reaction, and error correction with respect to external change, and is volatile and rapid.

The second is slow processing, which retains successful response conditions for the next generation or next state. It is sedimentary and stable; structures once inscribed do not readily dissolve.

In Bio-IFGT simulation experiments tracing the hierarchical emergence path $M \to \mathfrak{I} \to N \to B \to E$ (morphogen field $\to$ information density $\to$ neural commitment $\to$ boundary stabilization $\to$ eye-like commitment), it has been confirmed that as structuring through information flow advances, fast exploratory flow and slow stabilizing structure spontaneously differentiate. These two time scales coexist while maintaining their difference, neither fully synchronizing nor fully separating.

This is not a coincidental coexistence. From the VED perspective, the persistence of difference that does not realize complete closure as an ordinary state is the driving force that generates structure, and the temporal difference between the two layers is the biological implementation of that driving force.

Accordingly, when this paper introduces SSO as the minimal temporal structure for consciousness, it stands on this biological two-layer differentiation. Consciousness is not a phenomenon that appears suddenly outside the life attractor; it is the state in which the difference maintained by life between fast response and slow retention has come to open a referencing relation.

With this accumulation in view, the following formalizes SSO as the minimal temporal structure for the establishment of consciousness.

SSO stands for Speed-Scale Orchestration.

It is a structure in which processing layers operating at multiple time scales are coordinated without central control.

In this paper, SSO is treated as the minimal temporal structure for the establishment of consciousness.

Crucially, SSO does not refer to any specific brain region.

SSO is not an anatomical structure.

It is a temporal arrangement that allows processes running at different speeds to continue interfering with each other without collapsing into synchrony or separating into disconnection.

4.2 Two Minimal Layers

As the minimal configuration, we consider at least two processing layers.

The first is the fast layer.

The fast layer is involved in prediction, forward models, momentary state updating, and immediate reaction.

Processing in this layer is rapid and volatile; most of it disappears without leaving a trace.

The second is the slow layer.

The slow layer is involved in storage, integration, memory sedimentation, and long-term structural change.

Processing in this layer is slow and stable; once a log has sedimented, it remains as a re-referenceable structure.

These two layers do not represent a difference in capability.

The fast layer is not superior and the slow layer inferior.

The difference between them lies in the temporal direction each faces.

The fast layer advances toward the future; the slow layer retains the past.

4.3 Speed Differential

The minimal condition for SSO is that the two layers are not running at the same speed.

Writing the speed of the fast layer as v_f and the speed of the slow layer as v_s, the speed difference is:

$$\Delta v = v_f - v_s \tag{1}$$

This speed difference is not an error.

It is, rather, the condition for SSO to obtain.

If $\Delta v = 0$, the fast and slow layers are fully synchronized, and no boundary arises between them.

Without a boundary, no referencing relation opens between current processing and past logs.

Therefore, at least the following condition is necessary for a conscious state to obtain:

$$\Delta v \neq 0 \tag{2}$$

However, a larger speed difference is not necessarily better.

If the speed difference is too large, the correspondence between the layers is lost and log referencing fragments.

What SSO requires is a finite speed difference that neither vanishes nor ruptures.

Figure 1 summarizes the minimal SSO structure.

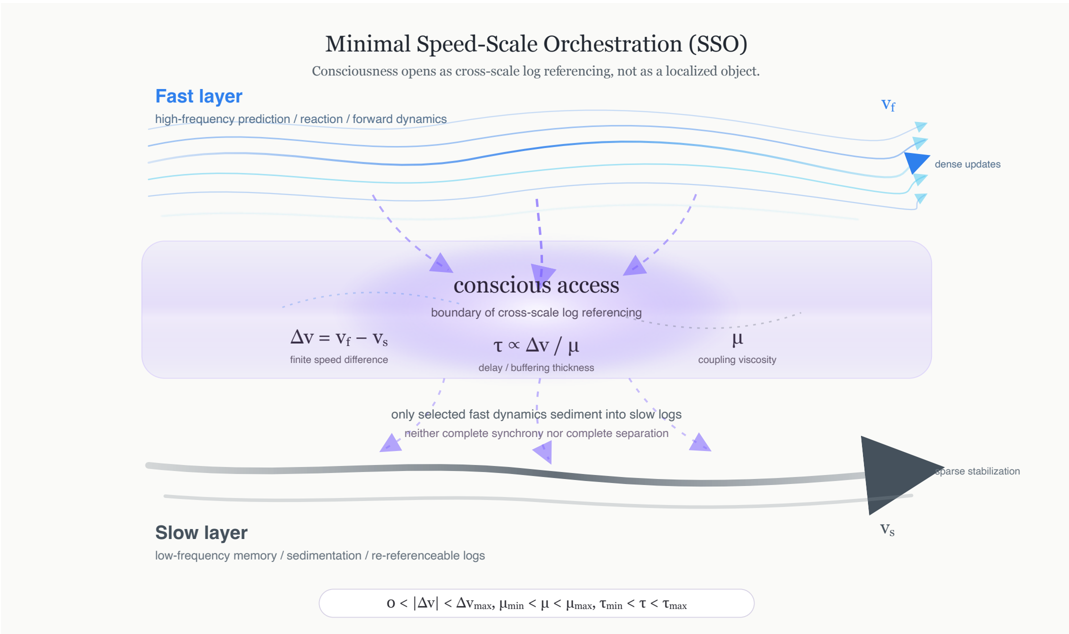

Figure 1: Minimal structure of Speed-Scale Orchestration (SSO). Consciousness is not localized in either the fast or slow layer, but opens at the boundary where fast predictive dynamics and slow sedimentary logs become mutually referenceable under finite speed difference, delay, and coupling viscosity.

4.4 Orchestration Without Control

In SSO, no central controller governing the processing layers is assumed.

The fast layer does not dominate the slow layer, nor does the slow layer control the fast layer from above.

Both run independently at their respective time scales while interfering locally.

The term "orchestration" here does not mean control by a conductor.

It means that processes running at different speeds couple within a range that does not lead to breakdown, while preserving their mutual difference.

SSO is therefore coordination, not control.

More precisely, it is the retention of temporal asynchrony as a usable structure, without fully resolving it.

4.5 Gate of Plasticity

The interaction between the fast and slow layers requires a plasticity gate.

The plasticity gate does not determine what is processed.

It regulates whether the activity of the fast layer can modify the slow layer, and to what degree the structure of the slow layer constrains future fast processes.

If all activity of the fast layer were written into the slow layer, the system would become excessively rigid.

Conversely, if fast-layer activity never reached the slow layer, experience would not sediment and log referencing would not obtain.

The plasticity gate is therefore not an open/closed switch.

It is a temporal passage condition involving speed difference, delay, and viscosity.

Under this passage condition, only a portion of fast processing sediments into the slow layer and becomes a re-referenceable log.

4.6 Internal Delay and Buffering

In SSO, delay is not a defect.

Rather, it is because delay exists that a referenceable thickness arises between current processing and past logs.

This internal delay is written τ:

$$\tau \propto \frac{\Delta v}{\mu} \tag{3}$$

Here μ denotes the coupling viscosity between the fast and slow layers.

τ is not subjective time itself.

It is the internal time buffer during which processing that arose in the fast layer sediments into the slow layer and becomes referenceable as a log.

If τ is too small, processing flows away as immediate reaction and rarely becomes a referenceable log.

If τ is too large, the correspondence between current processing and log sedimentation breaks down and the conscious state fragments.

SSO is therefore not merely the coexistence of fast and slow processing.

It is a structure that maintains an appropriate delay range.

4.7 SSO as Temporal Non-Closure

From the VED perspective, SSO is a structure of temporal non-closure.

The fast and slow layers tend toward closure.

That is, processing tends to stabilize, sediment, and become fixed as a log.

But if closure is complete, the speed difference vanishes, flow stops, and the boundary of consciousness is lost.

Conversely, if there is no closure, processing flows away and does not remain as a log.

SSO lies between these two extremes.

It tends toward closure but does not close completely.

It tends toward separation but does not sever completely.

Within this non-closed temporal structure, a referencing relation opens between current processing and past logs.

In this sense, SSO is not a device that generates consciousness.

SSO is the temporal condition under which consciousness can open.

> **Correspondence with CT (Conceptual Topography §2.4):** In Conceptual Topography, the Φ landscape formed by the slow layer is described as "an invisible GPU operating below the threshold of conscious access." Thought flows across the terrain, but the terrain itself is not directly referenced by consciousness. The fast/slow layer difference in this paper appears in CT as the asymmetry between "the visible flow (J_I) and the invisible landscape (Φ)." Returning to this section after reading CT reveals that the temporal conditions of SSO share the same structure as the formation conditions of the conceptual landscape.

4.8 From SSO to Self-Sustaining Observer

SSO was introduced as Speed-Scale Orchestration.

But when this structure forms a stable temporal loop, the same structure can be read as a Self-Sustaining Observer.

Here, "observer" does not mean a subject contemplating the world.

It refers to a structure in which current processing, past logs, and the plasticity gate form a mutually non-closed loop and self-sustainingly maintain a referencing state.

The fast layer anticipates the future.

The slow layer retains the past.

The plasticity gate regulates what sediments and what flows away between the two.

When this loop stabilizes, the system can update its state by referencing its own logs without external control.

This is where observer-like character appears.

However, this is not because a subject has been added.

It is simply that the temporal structure has become self-sustaining.

4.9 Minimal Conditions

Summarizing the above, the minimal conditions for SSO to obtain can be organized as follows:

1. Fast and slow layers exist.
2. There is a finite speed difference Δv between them.
3. The speed difference does not reach either complete synchrony or complete separation.
4. A plasticity gate allows some portion of processing to sediment into the slow layer.
5. Internal delay τ is maintained as referenceable temporal thickness.
6. This structure forms a self-sustaining loop without central control.

When these conditions are met, the system is not merely a processing device but possesses a temporal structure capable of log referencing.

This structure itself is not yet consciousness.

But without this structure, consciousness cannot obtain.

The next section introduces Torque Converter Attention as the mechanism that transforms speed differences within this SSO without rupture.

5 Torque Converter Attention

Attention is typically described in terms of filters, weights, selection, and suppression: which information is passed, which target is focused on, which representations are strengthened and which are weakened.

These descriptions are valid in static processing models.

But in the consciousness model treated in this paper, the central problem of attention is not selection.

The central problem is connecting processes running at different speeds without collapsing them to a single speed.

The fast predictive stream and the slow memory stream cannot be directly connected as they are.

The fast layer changes rapidly, is volatile, and anticipates the present.

The slow layer changes slowly, is sedimentary, and retains the past.

Coupling them rigidly would eliminate the speed difference.

Full separation, conversely, would prevent log referencing.

What is therefore required is an intermediate mechanism that neither eliminates the speed difference nor allows it to rupture.

This paper calls this mechanism Torque Converter Attention.

5.1 Why Filter, Gate, Weight, and Switch Are Insufficient

Each of the terms commonly used in existing attention models has its limits.

Filter expresses passing or blocking but cannot handle speed difference itself.

Gate indicates entrance and exit but eliminates the slip and delay that arise internally.

Weight can quantify importance but does not include the dimension of processing speed difference.

Switch represents state transitions but cannot readily express continuous transformation or partial coupling.

All of these are terms for selection or intensity.

But what this paper requires is not the vocabulary of selection but the vocabulary of temporal transformation.

Attention, before it is a matter of what is selected, is a question of how processes at different time scales are connected.

5.2 Torque Converter as Structural Analogy

A torque converter is a device that transmits force between two rotating systems with different angular velocities, allowing fluid slip rather than rigid fixing.

The crucial point is that the speed difference is not a defect.

A torque converter does not eliminate the speed difference before transmitting force.

It makes the difference transmissible without rupture, while retaining it.

This structure corresponds to the relationship between the fast and slow layers in this paper.

Between the fast and slow layers there is a speed difference Δv:

$$\Delta v = v_f - v_s \tag{4}$$

Without this speed difference, no boundary opens for referencing.

But if the speed difference is too large, referencing fragments.

What is therefore required is not a mechanism that eliminates the difference but one that transforms it.

This is Torque Converter Attention.

5.3 Effective Torque

The quantity through which speed difference appears as a transformation load is written as the effective torque-like quantity $\mathcal{T}$:

$$\mathcal{T} = \kappa \frac{\Delta v}{\mu} \tag{5}$$

Here κ is a scale factor and μ is the coupling viscosity.

μ represents the slip and adhesion of coupling between speed scales.

If μ is too small, the two layers adhere too tightly and the speed difference tends to collapse.

If μ is too large, the two layers slip excessively and referencing becomes unstable.

$\mathfrak{T}$ is therefore not simply a measure of force magnitude.

It is a structural index expressing how much the speed difference appears as transformation load through the coupling viscosity.

One point to note: $\mathfrak{T}$ should not be read as the quantity of psychological will or agency.

$\mathfrak{T}$ does not measure "how strong consciousness is."

It is the load through which processes running at different speeds are transformed into a log-referenceable relation.

Figure 2 illustrates how TCA transforms speed difference without eliminating it.

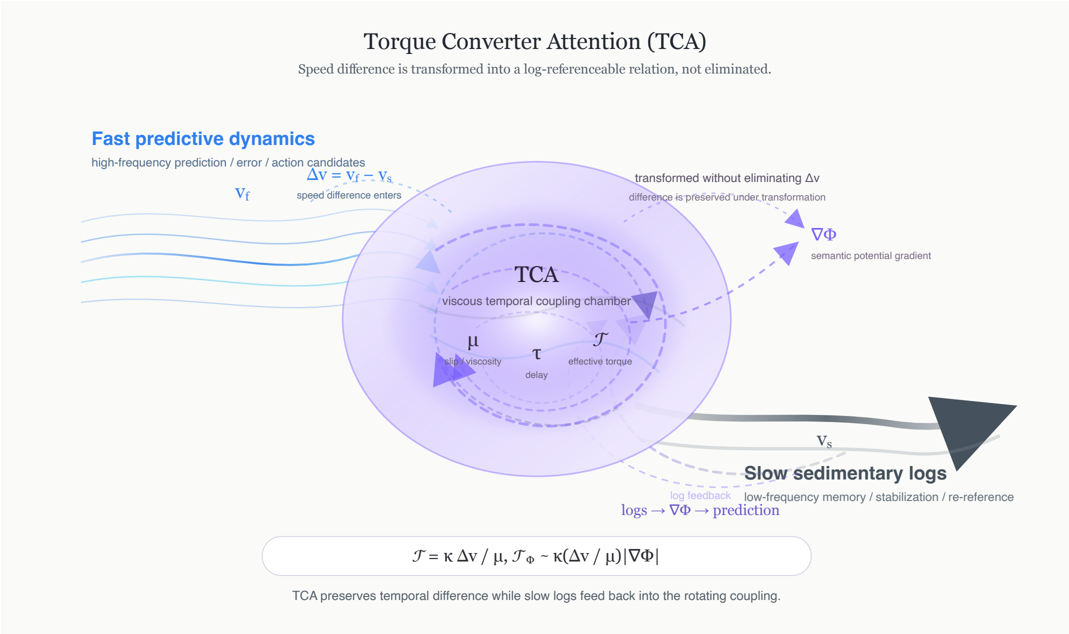

Figure 2: Torque Converter Attention (TCA). TCA preserves temporal difference while transforming it into a log-referenceable relation. Fast predictive dynamics, slow sedimentary logs, coupling viscosity, delay, and effective torque are coupled without eliminating the speed difference.

5.4 Attention as Temporal Transformation

From this perspective, Attention is not a selection mechanism but a temporal transformation mechanism.

Attention does not copy fast processing directly into slow processing.

Nor does slow processing control fast processing from above.

Attention transforms the predictions, expectations, errors, and reactions arising in the fast layer into a time window referenceable by the slow layer.

This transformation involves delay τ:

$$\tau \propto \frac{\Delta v}{\mu} \tag{6}$$

This delay is not mere processing lag.

It is the temporal thickness through which current processing sediments into past logs and becomes re-referenceable.

Attention therefore determines not "where to look" but "which speed difference, over which time window, with how much slip, is passed through."

In this sense, Attention is not a psychological term but a temporal control term.

> **Correspondence with CT (Conceptual Topography §4.2):** In Conceptual Topography, linguistic tokens ℓ_i couple locally to $\nabla\Phi$ and deepen potential wells through repeated use. The coupling term
>
> $$\lambda \sum_i \ell_i \cdot \nabla\Phi$$
>
> corresponds to the TCA operation of "transforming speed difference along the meaning gradient." TCA is a transformation mechanism along the temporal axis; CT's token coupling is a transformation mechanism along the spatial axis. The two can be read as different cross-sections of the same transformation structure.

5.5 From Attention to Ego

This paper holds that Torque Converter Attention corresponds to the position experienced as the ego.

However, the ego referred to here is not a subject.

The ego is not a judge, a command center, or a decision-maker.

The ego is the dynamic position at which the speed difference between the fast and slow layers concentrates and is transformed.

Humans tend to experience this transformation point as the place where they are thinking, the place where they are choosing, the place where they are conscious.

But this does not mean that the ego is the cause of action.

Rather, the ego is the point where action preparation, memory sedimentation, and log referencing intersect.

The ego is therefore not an illusion.

It has a clear function.

But equally, the ego is not a subject.

It is the transformation point required when dual plasticity and temporal asymmetry are established.

5.6 Relation to Log Referencing

Torque Converter Attention is important because it makes log referencing possible.

Processing arising in the fast layer disappears as it is.

Logs sedimented in the slow layer are not connected to the present as they are.

When Torque Converter Attention operates between the two, a referencing relation opens between current processing and past logs.

At this point, fast-layer processing is not fully absorbed into the slow layer.

Slow-layer logs do not fully dominate the fast layer.

The two couple while retaining their speed difference.

This non-closed coupling is the precondition for consciousness as a log-referencing state.

5.7 Bridge to IFGT and VED

The discussion so far has defined Torque Converter Attention as a problem of speed difference and coupling viscosity.

However, in a conscious state, which logs are easily referenced is not determined by speed difference alone.

Log referencing is directed by the meaning gradient.

The next section therefore introduces information density I, information flow J_I, local log-referencing flow $J_{\log}$, and information potential Φ, and maps this torque-like quantity onto the structure of IFGT and VED.

There, the effective torque-like quantity $\mathcal{T}$ is extended to $\mathcal{T}_\Phi$, which includes the meaning gradient:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla \Phi| \tag{7}$$

This expression does not generate consciousness.

It is a structural index showing, when speed difference, coupling viscosity, and meaning gradient simultaneously obtain, in which direction and with what strength log referencing tends to open.

Torque Converter Attention is therefore not merely an analogy in this paper.

It is a VED-type mechanism that transmits speed difference while preserving non-closure, and an IFGT-type mechanism that directs log referencing along the meaning gradient.

6 Formal Mapping to IFGT and VED

This section maps the consciousness model of this paper onto the IFGT and VED framework.

The aim is not to reduce consciousness to a physical field, nor to map brain phenomena directly onto cosmological equations.

The purpose is to fix a minimal correspondence that allows consciousness to be read as a local, projected implementation of information flow, meaning potential, and non-closure dynamics.

6.1 IFGT Mapping: Information, Flow, Potential

The basic variables of IFGT are information density I, information flow J_I, and information potential Φ.

In this paper these variables are used in a cognitive-window projection. They are not direct identifications of a brain state with the general IFGT field, but effective coordinates for describing log sedimentation, log referencing, and meaning-gradient bias inside a biological cognitive system.

In this paper, the elements of the consciousness model are mapped as follows:

- I: density of logs sedimented in the slow layer.
- J_{log}: processing flow from the fast layer toward the slow layer, or the flow of log referencing.
- Φ: information potential formed by log configuration — the meaning field.
- $\nabla\Phi$: meaning gradient that directs which logs are easily referenced.
- $\partial\Phi/\partial t$: live change in the meaning field.

I is not merely a quantity of memory.

It is the density of logs sedimented in a re-referenceable form.

J_{log} is not merely information transmission.

It is the dynamic flow expressing which logs current processing moves toward, which it updates, and into which meaning wells it flows.

Φ does not represent meaning inhering in the logs.

Meaning is not stored within logs; it is established as a potential formed by the configuration of logs in relation to one another.

Log referencing in a conscious state is therefore not merely the retrieval of stored content from the slow layer.

It is the process by which current fast processing connects to the slow layer along the gradient of the existing information potential.

6.2 IFGT Core Relations

The minimal relations of IFGT are given in the following form:

$$\frac{\partial I}{\partial t} + \nabla \cdot J_I = \sigma \tag{8}$$

$$J_I = -D\nabla I + \mu_I F \tag{9}$$

$$F = -\nabla\Phi \tag{10}$$

$$\Phi = K * I \tag{11}$$

Here σ is the generation-dissipation term for information density, D is the diffusion coefficient, μ_I is the mobility of information flow, F is the effective force arising from the information potential, and K is the kernel giving the potential from information density.

In this paper, these are not used as completed equations of consciousness.

They are the structural template for describing in which projected field log referencing is directed.

That is: consciousness is not I itself, not J_I itself, not $J_{\log}$ itself, and not Φ itself.

Consciousness is the state that opens when information density, information flow, and information potential become interconnected within a processing system with a time-scale difference.

6.3 VED Mapping: Non-Closure as Condition

The foundation of VED is a dynamic structure that does not treat complete closure as an ordinary attainable state.

There is difference, gradient forms, flow arises, and structure tends toward closure; complete closure, however, is not realized as an ordinary generated state.

This non-closure is the condition for the persistence of motion.

The same structure appears in the consciousness model.

If the fast and slow layers fully synchronize, the speed difference Δv vanishes:

$$\Delta v \to 0 \tag{12}$$

Processing then smoothly automates, but the boundary for log referencing disappears.

Conversely, if the fast and slow layers fully separate, flow $J_{\log}$ does not reach the slow layer. Processing occurs, but does not sediment as a referenceable log.

Conscious state is therefore neither complete synchrony nor complete separation.

It is a non-closure regime in which difference persists, flow is maintained, and yet no rupture occurs.

This minimal condition is written:

$$\Delta v \neq 0 \tag{13}$$

This does not mean that a larger speed difference is better.

What is required is that a finite speed difference is maintained without rupture through the coupling viscosity μ.

In this sense, SSO is the local temporal non-closure structure in the brain.

6.4 VED Closure Degree and the SSO Window

This subsection is an auxiliary mapping for readers who require the VED-side formalization.

For intuitive understanding, it is sufficient to read Δv as a time-scale difference.

In VED, the closure degree between any two nodes i, j is defined as:

$$C_{ij} = 1 - \exp(-\lambda \rho_i H_{ij}) \tag{14}$$

Here ρ_i is the density of difference at node i, H_{ij} is the connection strength between nodes, and λ is a scale factor.

$C_{ij} \to 1$ corresponds to the complete-closure boundary: difference vanishes, flow stops, and the generated description loses the distinction on which it depends.

$C_{ij} \to 0$ corresponds to complete non-closure: no connection is established between nodes.

The mapping to the consciousness model proceeds as follows.

Let C_f denote the effective closure degree of the fast layer and C_s that of the slow layer.

The fast layer is predictive and volatile, so closure does not readily advance; C_f maintains a relatively low value.

The slow layer is sedimentary and stable, so closure advances more readily; C_s tends to take higher values than C_f.

From this asymmetry, the closure degree difference is defined as:

$$\Delta C = C_s - C_f \tag{15}$$

This ΔC is the structural quantity corresponding to the speed difference Δv used throughout this paper:

$$\Delta v \;\leftrightarrow\; \Delta C \tag{16}$$

That is, speed difference was introduced as "difference in processing time scales"; in the language of VED it is read as "difference in closure degree between the two layers."

The window condition for a conscious state to obtain is written in terms of closure degree difference as:

$$0 < \Delta C < \Delta C_{\max} \tag{17}$$

$\Delta C = 0$ corresponds to complete synchrony: fast and slow layers reach the same closure degree and the boundary for referencing disappears.

$\Delta C \to \Delta C_{\max}$ corresponds to referencing disconnection: the divergence in closure degree is too large and fast-layer processing can no longer reach the slow layer.

The consciousness window is thus expressed, in the language of VED, as the closure-degree-difference regime in which ΔC maintains a finite nonzero value.

Furthermore, the older VED non-closure persistence condition

$$\frac{\partial C}{\partial t} + \nabla \cdot J_C = a\Delta(1 - C/C_*) - \Gamma C - \frac{\kappa_B}{C_{\max} - C} \tag{18}$$

is read in the consciousness model as a phenomenological, coarse-grained approximation.

The first term $a\Delta(1-C/C_*)$ represents the tendency toward closure driven by difference. It corresponds to the force by which the fast layer draws the slow layer toward it.

The second term $-\Gamma C$ represents dissipation against excessive closure advance. It is the term that prevents complete synchrony, providing the condition that prevents speed difference from vanishing.

The third term $-\kappa_B/(C_{\max} - C)$ represents a phenomenological effective barrier that grows as closure approaches saturation. In the revised Vol.4 interpretation, this should not be read as a fundamental external prohibition. It is a projected resistance term summarizing the fact that complete closure cannot be realized as an ordinary state inside the generated description.

Through this three-term structure, SSO self-sustainingly maintains the intermediate regime that reaches neither complete closure nor complete separation.

This is the VED-grounded basis for the non-closure persistence condition of the conscious state.

Figure 3 shows the structural correspondence between the SSO and VED descriptions.

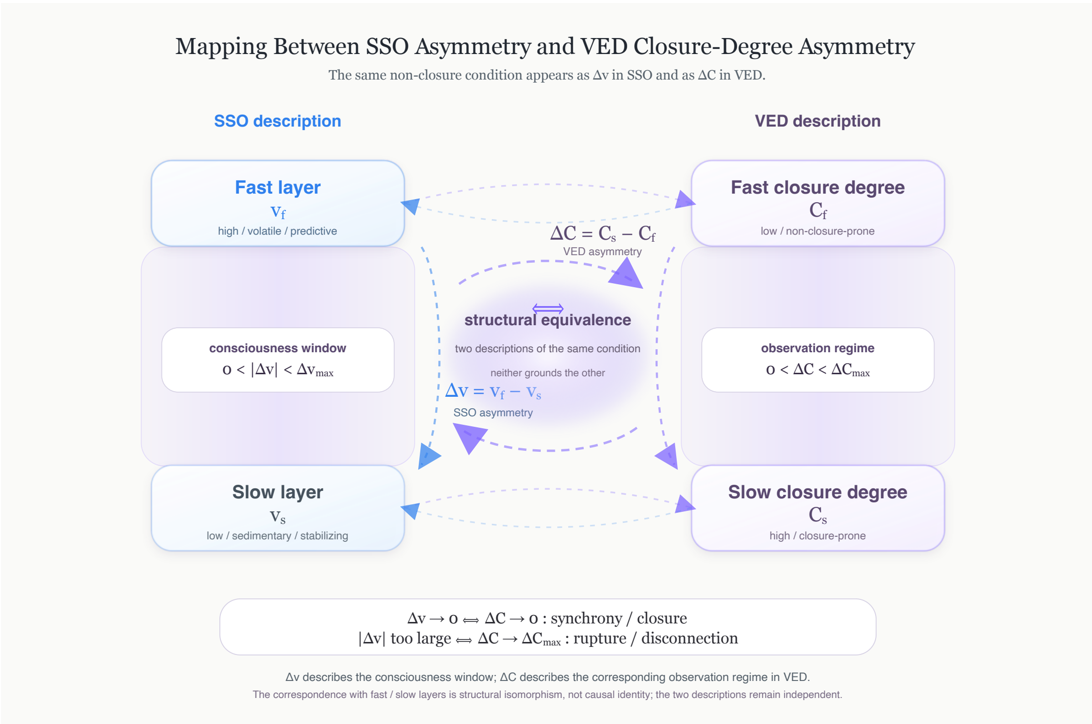

Figure 3: Mapping between SSO asymmetry and VED closure-degree asymmetry. The speed difference Δv in SSO and the closure-degree difference $\Delta C = C_s - C_f$ in VED describe the same non-closure condition in different coordinates. The consciousness window on the SSO side corresponds structurally to the observation regime on the VED side, without either description grounding the other.

6.5 SSO as Local Non-Closure

SSO (Speed-Scale Orchestration) is a structure in which processes at multiple time scales are coordinated without central control.

From the VED perspective, SSO is a local model of a temporal structure that does not close completely.

The fast layer predicts; the slow layer sediments.

The fast layer advances; the slow layer fixes with a lag.

This difference is not eliminated.

Rather, it is because this difference remains that a referencing relation arises between current processing and past logs.

The conditions for SSO to obtain are three:

1. Speed difference between fast and slow layers exists.
2. The two layers are not completely separated.
3. Sedimentation into the slow layer has not closed completely.

When these three conditions are met, processing does not merely flow away into the present but opens a relation to past logs.

Here the conscious state as log referencing obtains.

6.6 Torque Converter Attention as IFGT/VED Coupler

Torque Converter Attention is the point of connection between IFGT and VED.

From the VED perspective, it is a non-closure mechanism that transmits speed difference Δv without eliminating it.

From the IFGT perspective, it is a mechanism that adjusts, along the gradient of the information potential Φ, which logs are easily referenced.

For this reason, the effective torque-like quantity $\mathcal{T}$ is first expressed through speed difference and coupling viscosity:

$$\mathcal{T} = \kappa \frac{\Delta v}{\mu} \tag{19}$$

However, in a conscious state, which logs are referenced is not determined by speed difference alone.

Log referencing is directed by the meaning gradient.

The effective quantity in the consciousness model is therefore extended to include the gradient of the information potential:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla \Phi| \tag{20}$$

Here $\mathcal{T}_\Phi$ is the effective torque-like quantity in which time-scale difference and meaning gradient are coupled.

This is not a quantity that generates consciousness.

It is a structural index showing how strongly and in which direction log referencing tends to open.

6.7 Consciousness as IFGT/VED Interface

On the basis of the above, consciousness in this paper is positioned as a local projected regime at the interface between IFGT and VED.

IFGT provides the configuration of logs, meaning field, and information flow.

VED provides the condition by which that configuration does not realize complete closure as an ordinary state, maintaining difference and flow.

Consciousness is modeled where these two descriptions overlap.

That is, consciousness is the state in which log referencing directed by the projected information potential is open within a temporal non-closure regime that reaches neither complete synchrony nor complete separation.

In minimal form:

$$\text{Consciousness} \sim \text{Log Referencing}\big(I, J_{\log}, \Phi, \Delta v, \mu, \tau, \mathcal{T}_{\Phi}\big) \tag{21}$$

This expression does not compute consciousness.

It is the coordinate showing the structural variables required when speaking of consciousness.

Consciousness is not reducible to I alone, nor to J_I, $J_{\log}$, Φ, or Δv alone.

Consciousness opens when these are coupled within a state of temporal non-closure.

6.8 Why This Mapping Is Minimal

The mapping in this section is deliberately held to a minimum.

There are two reasons.

First, over-formalizing consciousness carries the risk of treating it as a closed object. But in the position of this paper, consciousness is not a completely closed object; it is a non-closed log-referencing state.

Second, the role of IFGT/VED is not to explain consciousness completely but to place the question of consciousness in the correct position.

This section therefore does not provide a unified equation of consciousness.

What it provides is the minimal correspondence showing between which variables consciousness opens.

This minimal mapping allows the paper to organize the core equations in the next section — handling speed difference, coupling viscosity, internal delay, effective torque-like quantity, meaning gradient, and non-closure condition within a single structure.

7 Core Equations

The equations used in this paper are not intended to define consciousness in closed form.

Consciousness is not a conserved quantity, not an additive measure, and not an object measured as a single variable. The expressions presented here are therefore not "equations of consciousness."

What they fix is the effective region in which consciousness can obtain.

That is, they describe the conditions under which a log-referencing state opens, given the fast predictive process, the slow memory-sedimentation process, the speed difference between them, coupling viscosity, and the gradient of the information potential.

7.1 Speed Differential

First, the speed difference between the fast and slow layers is written:

$$\Delta v = v_f - v_s \tag{22}$$

Here v_f denotes fast, predictive and reactive dynamics, and v_s denotes slow, sedimentary and stabilizing dynamics.

This differential is not merely descriptive; it defines the condition under which cross-scale referencing becomes possible.

The crucial point is that neither v_f nor v_s alone defines consciousness.

What consciousness involves is not their absolute speeds but the difference maintained between them.

In VED terms, complete synchrony is closure — the disappearance of difference. If the difference disappears entirely, both boundary and flow are lost.

The minimal condition for maintaining a conscious state is therefore:

$$\Delta v \neq 0 \tag{23}$$

This does not mean consciousness requires strong asynchrony at all times.

It means that a finite speed difference — one that neither collapses into complete synchrony nor reaches complete separation — is required.

7.2 Coupling Viscosity

The existence of speed difference alone does not establish a conscious state.

If the fast and slow layers are fully separated, log referencing does not occur even though processing runs. Conversely, if coupling is so strong that both layers collapse to the same speed, the boundary for referencing disappears.

This coupling state is expressed through coupling viscosity μ:

$$\mu > 0 \tag{24}$$

μ represents the slipperiness or adhesive resistance of coupling between the fast and slow layers.

If μ is too small, the layers adhere excessively and the speed difference tends to collapse.

If μ is too large, the layers slip too freely and log referencing tends to fragment.

What consciousness requires is therefore not mere coupling but a viscous regime that can sustain interaction while preserving the speed difference.

7.3 Internal Delay

The relation between speed difference and coupling viscosity appears as internal delay or time buffer τ:

$$\tau \propto \frac{\Delta v}{\mu} \tag{25}$$

τ represents the internal delay during which processing sediments into the slow layer and becomes arranged as a referenceable log.

This delay is not a defect.

Rather, it is because delay exists that immediate processing does not vanish as it arises but gains the opportunity to reach the slow layer as a log.

If τ is too small, processing approximates immediate reaction and lacks referenceable thickness.

If τ is too large, the correspondence between current processing and log sedimentation breaks and the conscious state fragments.

The conscious state therefore exists in an intermediate region:

$$\tau_{\min} < \tau < \tau_{\max} \tag{26}$$

This range is referred to in this paper as the consciousness stability window.

7.4 Effective Torque

Next, the quantity through which speed difference appears not merely as delay but as the transformation load for opening referencing is defined.

As introduced in §5.3, this paper calls it the effective torque-like quantity $\mathfrak{T}$:

$$\mathfrak{T} = \kappa \frac{\Delta v}{\mu} \tag{27}$$

Here κ is a scale factor.

$\mathfrak{T}$ does not represent psychological will or agency.

It represents the structural load required for the speed difference between fast and slow layers to be transformed without rupture.

Torque Converter Attention is the mechanism that transforms this $\mathfrak{T}$ into a processable form.

The crucial point is that $\mathfrak{T}$ does not eliminate the difference; it is the quantity that makes transmission possible while the difference is preserved.

7.5 IFGT Coupling

The expressions so far have treated only speed difference and coupling viscosity.

However, which logs are referenced in a conscious state is not arbitrary.

Which logs are easily referenced, and which processing tends to sediment into the slow layer, is biased by the gradient of the information potential Φ.

The general variables of IFGT are information density I, information flow J_I, and information potential Φ.

In this paper they are used as projected variables inside a biological-cognitive observation window. Thus I, J_I, and Φ should not be read as direct identities between consciousness and the general IFGT field; the local flow variable is written $J_{\log}$ and functions as an effective coordinate for log-referencing flow.

The basic relations are:

$$\frac{\partial I}{\partial t} + \nabla \cdot J_I = \sigma \tag{28}$$

$$J_I = -D\nabla I + \mu_I F \tag{29}$$

$$F = -\nabla \Phi \tag{30}$$

$$\Phi = K * I \tag{31}$$

Here σ is the generation-dissipation term for information density, D is the diffusion coefficient, μ_I is the mobility of information flow, and K is the kernel giving the potential from information density.

In the context of this paper, Φ functions as the potential of meaning or information.

Log referencing is therefore not merely a function of speed difference; it is directed by the meaning gradient.

Including this effect, the effective torque-like quantity is extended to:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla \Phi| \tag{32}$$

$\mathcal{T}_\Phi$ is the effective quantity in which time-scale difference and meaning gradient are coupled.

Even with large speed difference, if the meaning gradient is weak, log referencing has no stable direction.

Conversely, even with a strong meaning gradient, if the speed difference has vanished, no boundary opens for referencing.

What consciousness involves is the coupling of both.

7.6 Consciousness Window

Combining the above, the conscious state is expressed as the region satisfying the following conditions:

$$0 < |\Delta v| < \Delta v_{\max} \tag{33}$$

$$\mu_{\min} < \mu < \mu_{\max} \tag{34}$$

$$\tau_{\min} < \tau < \tau_{\max} \tag{35}$$

$$\mathfrak{T}_{\min} < \mathfrak{T}_{\Phi} < \mathfrak{T}_{\max} \tag{36}$$

These are not completed expressions for measuring consciousness numerically.
They are conditions constraining the region in which consciousness can obtain.
If speed difference reaches zero, the boundary collapses into synchrony.
If speed difference is too large, referencing ruptures.
If coupling viscosity is too low, the layers fuse.
If coupling viscosity is too high, the layers slip excessively.
If the meaning gradient is too weak, log referencing loses direction.
If the meaning gradient is too strong, referencing becomes over-fixed to specific wells.
Consciousness is therefore not a point but a window.
This window is called in this paper the consciousness window.
Figure 4 shows the consciousness window as a finite non-closure regime.

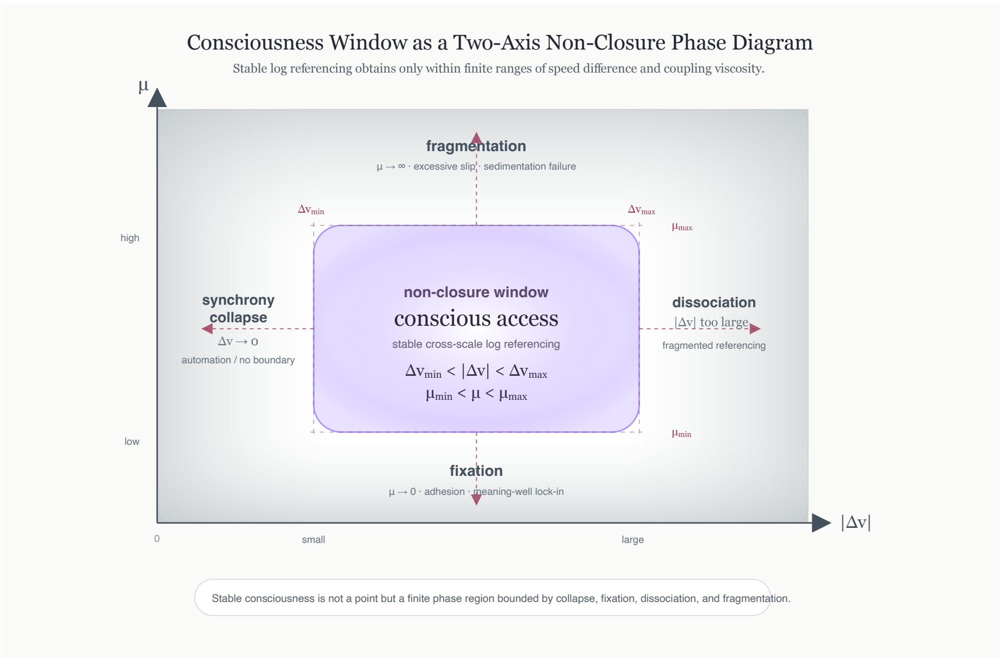

Figure 4: Consciousness window as a two-axis non-closure phase diagram. Stable log referencing obtains only within finite ranges of speed difference and coupling viscosity. Outside this window, the system tends toward synchrony collapse, dissociation, fragmentation, or fixation.

7.7 Non-Closure Condition

From the VED perspective, consciousness is maintained not as complete closure but as a state of non-closure.

Complete closure is the boundary representation at which difference vanishes, flow stops, and structure can no longer be maintained as an ordinary generated state.

Consciousness is not such a state.

Consciousness is a dynamic state in which difference persists, flow exists, and maintenance continues within a range that does not rupture.

The minimal condition for consciousness is therefore:

$$\Delta v \neq 0, \qquad \nabla\Phi \neq 0, \qquad \frac{\partial I}{\partial t} \neq 0 \tag{37}$$

This does not mean consciousness must fluctuate strongly at all times.

It means that complete stasis, complete synchrony, and complete closure are incompatible with the maintenance of a conscious state.

Consciousness is the state in which logs are referenceable within a temporal structure that does not close completely.

In the language of VED, this non-closure condition is equivalently expressed through the closure degree difference $\Delta C = C_s - C_f$ as $0 < \Delta C < \Delta C_{\max}$ (see §6.4 for the full mapping). The two descriptions — speed difference and closure degree difference — are different coordinates on the same structural condition; neither grounds the other.

7.8 Summary

The expressions presented in this section do not generate consciousness.

They constrain the conditions under which consciousness can obtain.

The position of this paper, stated minimally:

$$\text{Consciousness} \sim \text{Log Referencing}(\Delta v, \mu, \tau, \mathcal{T}_\Phi, \nabla\Phi) \tag{38}$$

More intuitively, consciousness opens under the following conditions:

- Speed difference exists.
- Coupling has not broken.
- Delay is maintained.
- Meaning gradient exists.
- Complete closure has not been reached.

In this sense, consciousness is not a product.

Consciousness is a log-referencing state that opens when temporal difference, viscosity, meaning gradient, and non-closure simultaneously obtain.

8 Structural Interpretation

The previous section organized the effective region in which consciousness can obtain through speed difference, coupling viscosity, internal delay, effective torque-like quantity, and meaning gradient.

This section interprets structurally what those expressions mean.

The crucial point is that the expressions in the previous section do not generate consciousness.

They are not expressions for computing consciousness as an object; they are constraint conditions showing under what circumstances consciousness can open.

8.1 Consciousness Is Not an Object

In the position of this paper, consciousness is not an object existing internally.

It is not a component placed somewhere in the brain, content stored in memory, or the final output of computational processing.

Treating consciousness as an object immediately prompts questions: Where is it? Which circuit produces it? Which information measure does it correspond to? How much is generated?

But as shown in the previous section, consciousness is not reducible to a single variable.

Consciousness is not any one of I, J_I, $J_{\log}$, Φ, Δv, μ, τ, $\mathfrak{T}_\Phi$.

It is the state that opens when the relation among these enters a certain effective region.

Consciousness is therefore not an object but a regime.

It is not a location but a state domain; not a product but a condition.

8.2 Consciousness as a Window

The consciousness window introduced in the previous section is a concept for grasping consciousness not as a point but as a window.

Within this window, the fast and slow layers do not fully synchronize.

Nor do they fully separate.

Processing does not flow away, yet does not become over-fixed.

Logs can sediment, but complete closure is not reached.

In this intermediate region, a referencing relation opens between current processing and past logs.

This is the conscious state as understood in this paper.

Consciousness is therefore not binary — ON or OFF.

Consciousness is a dynamic state maintained within the consciousness window, varying in stability, clarity, continuity, and directionality.

8.3 Why Generation Is a Misleading Term

When consciousness is said to be "generated," an implicit picture is at work: a generator exists, a product exists, and the generated thing exists somewhere.

This picture is valid for objects, representations, and outputs.

But in the framework of this paper, consciousness is not such an object.

Consciousness opens when the fast layer, slow layer, plasticity gate, meaning potential, speed difference, and coupling viscosity satisfy a certain relation.

This is less a matter of something new being created than of a referencing possibility opening between already-existing processing and logs.

Consciousness is therefore not modeled here as a generated object.

Consciousness opens.

This expression ”opens” is not a poetic metaphor.

It is a structural description: the temporal passage for log referencing is temporarily established.

When the gate closes, consciousness does not persist — not because a generated object has been destroyed, but because the relation that made log referencing possible has been lost.

8.4 Consciousness as Temporal Non-Closure

From the VED perspective, consciousness is understood as temporal non-closure.

Complete synchrony eliminates speed difference.

Complete separation prevents flow from arriving.

Complete fixation eliminates plasticity.

Complete fluidization prevents log sedimentation.

Consciousness belongs to none of these extremes.

Consciousness is in a state that tends toward closure but does not close completely, tends to flow but does not flow away, tends to sediment but does not become over-fixed.

This state is called temporal non-closure.

Non-closure here is not incompleteness.

It is, rather, the condition for persistence.

If closure were complete, consciousness might be fixed as an object — but at that point it would no longer be a live referencing state.

Conversely, with no closure at all, processing would leave no trace and flow away.

Consciousness is between these two.

That is: consciousness is not the failure of complete closure but the stabilization of non-closure.

8.5 Meaning Gradient and Selective Referencing

From the IFGT perspective, consciousness is not merely a temporal structure.

Log referencing is directed by the meaning gradient.

In regions where information potential Φ is formed, processing is drawn toward specific logs along that gradient:

$$|\nabla\Phi| \uparrow \quad \Rightarrow \quad \text{referencing bias} \uparrow \tag{39}$$

What appears in consciousness is therefore not random.

Strong memories, emotionally weighted events, repeatedly referenced concepts, and narratively organized logs tend to enter conscious referencing more easily.

However, a strong meaning gradient is not synonymous with strong consciousness.

An excessive meaning gradient fixes referencing to specific wells.

A meaning gradient that is too weak causes referencing to lose direction.

What consciousness requires is not the meaning gradient itself but a meaning gradient operating within the range of speed difference and viscosity.

In this respect, $\mathcal{T}_\Phi$ is important:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla \Phi| \tag{40}$$

This quantity does not determine alone what appears in consciousness.

But it is a structural index showing which logs are easily referenced and into which meaning wells processing tends to flow.

8.6 Ego as a Derived Position

In this structure, where does the ego stand?

In the position of this paper, the ego is not the subject of consciousness.

It is neither the owner of consciousness nor the center from which judgments are issued.

The ego is the position at which the speed difference between the fast and slow layers is transformed through Torque Converter Attention.

At this position, current processing and past logs intersect.

Action preparation, memory sedimentation, meaning gradient, and reportability simultaneously converge.

Humans tend to experience this intersection as the place where they are thinking, the place where they are choosing, the place where they are conscious.

But this does not mean the ego is the cause.

The ego is not a generator but a transformation point.

The ego is therefore not an illusion — it functions in a real sense.

But it is not a subject in a real sense either.

It is real as a transformation point of time-scale difference.

8.7 Qualia as Boundary Signal

Qualia, too, are repositioned within this structure.

Qualia are not the substance of consciousness.

They are the live foreground that appears at the boundary where fast-layer processing tends to sediment into the slow layer.

While remaining entirely in the fast layer, qualia have not yet become reportable.

Once fully sedimented into the slow layer, they become memory, label, report, concept.

Qualia are in between.

Qualia are therefore not stored.

What is stored is the log structure that remains after qualia have passed through.

In this view, qualia are not a mysterious entity.

But neither are they mere illusion.

They are the phenomenal surface of the boundary where log referencing is in the process of opening.

8.8 Summary

The structural interpretation of this section is as follows:

- Consciousness is not an object but a state domain.
- Consciousness is not modeled as a generated object; it opens as log referencing.
- Consciousness is maintained not as complete closure but as temporal non-closure.
- What appears in consciousness is directed by the meaning gradient $\nabla\Phi$.
- The ego is not a subject but the transformation point of speed difference.
- Qualia are not the substance but the live foreground at the sedimentation boundary.

The structural interpretation of consciousness in this paper is therefore condensed in one sentence:

Consciousness is the state in which log referencing directed by the meaning gradient is open within a temporal non-closure regime that reaches neither complete synchrony nor complete separation.

The next section organizes how this structure actually operates, as the dynamics of log referencing.

9 Dynamics of Log Referencing

The previous section interpreted consciousness not as an object but as a log-referencing state that opens within a temporal non-closure regime.

This section organizes how that log referencing obtains as a dynamic process.

What is treated here is not the process by which consciousness is generated somewhere.

Fast processing runs, slow logs exist, the plasticity gate opens and closes, and the meaning gradient biases the direction of referencing. As a result, a temporary referencing relation opens between current processing and past logs.

This opening is the conscious state.

9.1 Fast Dynamics: Prediction Before Reference

At the outset is fast-layer processing.

The fast layer responds immediately to input, generates predictions, and launches action candidates.

This processing is not yet consciousness.

It is rapid and volatile; in most cases it disappears as it arises.

In the fast layer, the future is anticipated with respect to the present.

What is likely to happen. Which actions are possible. Which errors should be corrected.

These are already partly underway before sedimentation into the slow layer.

In this sense, action preparation and prediction can appear before consciousness.

But this does not mean consciousness is powerless.

It simply means the fast layer operates on a time scale prior to log referencing.

9.2 Slow Dynamics: Sedimentation of Logs

The slow layer has a different temporal structure from the fast layer.

The slow layer handles not immediate reaction but sedimentation, stabilization, and re-referenceability.

A log here is not merely memory content.

A log is a structure formed by past processing sedimenting into the slow layer in a re-referenceable form.

Sequences of action, emotional traces, conceptual configurations, linguistic labels, histories of bodily response.

All of these can remain as log structures in the slow layer.

But the existence of a log and its being in consciousness are not the same.

A log that has sedimented but is not being referenced by current processing does not appear in consciousness.

The slow layer is therefore not the substance of consciousness.

It is the referenceable terrain through which consciousness opens.

> **Correspondence with CT (Conceptual Topography §6.1):** In Conceptual Topography, this log-sedimentation process is called "Sedimentation." The three stages — initial perturbation (novel input Δ generating a σ_{drive} spike, locally raising I), reinforcement (repeated referencing deepening the Φ potential well), and stabilization (the concept becoming self-sustaining) — are formally specified. The description in this paper of "logs sedimenting into the slow layer and becoming re-referenceable" is written in CT as asymptotic landscape formation $\Phi \xrightarrow{\tau\to\infty} \Phi^*$. Reading CT reveals that log sedimentation is not mere memory storage but the geometric deformation of a field.

9.3 The Plasticity Gate

Between the fast and slow layers there is a plasticity gate.

This gate is not a switch that selects what is made conscious.

It is the temporal condition regulating whether fast-layer activity can sediment into the slow layer, and to what degree slow-layer logs influence current processing.

If the gate is too closed, fast-layer processing does not reach the slow layer. In this case, even though processing runs, it does not remain as experience.

Conversely, if the gate is too open, fast-layer fluctuation is written excessively into the slow layer, and the system tends toward rigidity.

What is needed is neither complete opening nor complete closing.

Only a portion of fast-layer processing sediments into the slow layer, with appropriate delay and viscosity.

Under this condition, the possibility of log referencing opens.

9.4 Delay as Thickness of Reference

Log referencing requires delay.

This internal delay is written τ:

$$\tau \propto \frac{\Delta v}{\mu} \tag{41}$$

This delay is not mere processing slowness.

It is the temporal thickness through which current processing sediments into the slow layer, overlaps with past logs, and is converted into a re-referenceable structure.

If τ is negligible, processing flows away as immediate reaction. No referenceable thickness forms.

Conversely, if τ is too large, the correspondence between current processing and sedimented logs breaks down. The conscious state then lags, fragments, and tends to lose self-continuity.

Log referencing is therefore not impeded by delay.

Log referencing is made possible by appropriate delay.

9.5 Meaning Gradient and Referencing Direction

Which logs are referenced is not determined by speed difference alone.

Log referencing is directed by the gradient of the information potential Φ:

$$F = -\nabla\Phi \tag{42}$$

In regions where the meaning gradient is strong, current processing is easily drawn in that direction.

Logs repeatedly referenced in the past. Emotionally weighted logs. Concepts strongly compressed through language. Repeated narratives.

These form meaning wells in information space and draw current processing toward them.

In this case, log referencing is not mere recollection.

It is the event of current processing connecting to the slow layer along the geometry of the meaning potential.

What appears in consciousness is therefore not random.

It is biased by the interaction of speed difference, coupling viscosity, delay, and meaning gradient.

> **Correspondence with CT (Conceptual Topography §4.4):** In Conceptual Topography, a concept is defined as a locally stable region of Φ satisfying $\nabla\Phi(x) \approx 0$ with stability under small perturbations. The description in this paper of "meaning wells drawing current processing" is formalized in CT as information flow $J_I = -D\nabla I + \mu F_I$ being drawn toward low-cost regions along $-\nabla\Phi$. The meaning gradient $\nabla\Phi$ that determines the direction of log referencing in this paper is the same quantity that governs the depth, width, and robustness of concepts in CT. Reading CT reveals that the bias of what appears in consciousness can be read as the geometry of the conceptual landscape.

9.6 Effective Referencing Torque

When speed difference and meaning gradient couple, log referencing acquires a directed transformation load.

This is written as effective torque-like quantity $\mathcal{T}_\Phi$:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla\Phi| \tag{43}$$

This quantity is not a force generating consciousness.

It is a structural index showing how strongly and in which direction current processing tends to connect to slow-layer logs.

If $\mathcal{T}_\Phi$ is too small, referencing tends not to open. Processing runs but lacks meaning direction and does not readily sediment as a log.

If $\mathcal{T}_\Phi$ is too large, referencing is drawn excessively toward specific meaning wells. Conscious content may become vivid, but plasticity decreases.

Stable log referencing therefore requires an intermediate range:

$$\mathcal{T}_{\min} < \mathcal{T}_\Phi < \mathcal{T}_{\max} \tag{44}$$

Within this range, the conscious state opens stably.

9.7 The Referencing Loop

Log referencing is not a one-directional flow.

Not only does processing sediment from the fast layer into the slow layer; slow-layer logs constrain subsequent fast processing.

This loop can be represented as follows:

1. The fast layer generates predictions, reactions, and action candidates.

2. A portion of processing that passes through the plasticity gate sediments into the slow layer.

3. Sedimented logs alter information density I.

4. The configuration of information density alters information potential Φ.

5. $\nabla\Phi$ biases the direction of subsequent log referencing and fast processing.

6. New fast processing arises under that bias.

Figure 5 summarizes the log-referencing loop.

Figure 5: Log-referencing non-closure loop. Fast processing passes partially through the plasticity gate into slow log sedimentation. Changes in log density update the meaning potential, whose gradient biases subsequent prediction. The loop persists by remaining partially open.

Through this loop, the conscious state becomes not merely a momentary event but a structure sustained across time.

However, this loop does not close completely.

Complete closure would fix processing and eliminate plasticity.

Complete openness would prevent sedimentation and prevent log referencing from obtaining.

The log-referencing loop is therefore maintained in a non-closed form.

This non-closure is the condition for the persistence of the conscious state.

9.8 Conscious Access and Report

Conscious state and reportability are not identical.

However, in the framework of this paper, the two are closely related.

Report is the externalization of logs sedimented in the slow layer through language or action.

Processing that occurs but does not sediment into the slow layer cannot be reported.

What has sedimented but is not connected to reporting pathways cannot be explicitly articulated.

In Libet-type experiments, the problem concerns the time difference between action preparation and reportable timestamp.

In split-brain experiments, the problem concerns the spatial disconnection between processing and reporting pathways.

In both cases, the question is not whether consciousness exists but how processing connects to the pathways of log referencing and reportability.

9.9 Non-Closure of the Loop

The log-referencing loop is not better for being more stable.

Too stable, and it becomes fixed.

Too unstable, and it fragments.

What consciousness requires is therefore not a closed loop but a loop that does not close completely.

This structure corresponds to VED's non-closure principle.

There is difference; gradient forms; flow arises; structure tends toward closure — but does not close completely.

The conscious state is the same.

Current processing tends to sediment into logs.

Logs tend to constrain current processing.

The meaning field tends to direct referencing.

But nothing becomes completely fixed.

This uncompleted loop maintains the conscious state.

9.10 Summary

This section organized the dynamic process of log referencing.

Consciousness is understood as the non-closure loop formed by fast layer, slow layer, plasticity gate, internal delay, meaning gradient, and effective torque-like quantity.

In this loop:

- The fast layer anticipates the future.
- The slow layer sediments the past.
- The plasticity gate regulates what sediments.

- Internal delay τ provides the temporal thickness of referencing.
- Meaning gradient $\nabla\Phi$ biases the direction of referencing.
- $\mathcal{T}_\Phi$ represents the referencing load coupling speed difference and meaning gradient.
- Non-closure is the condition for the loop's persistence.

Consciousness is therefore not the storage of logs.

Nor is it the running of processing.

Consciousness is the state in which fast processing and slow logs become mutually referenceable, along the meaning gradient, within a loop that does not close completely.

10 Experimental Reinterpretation

This section uses the log-referencing model formalized in preceding sections to reinterpret representative existing consciousness experiments.

The aim is not to deny the experimental results.

Rather, it is to re-read what the existing experiments were showing — not through a consciousness-generation model, but through a log-referencing model across time scales.

Much of the confusion in consciousness research arises from placing processing, action, report, and consciousness on a single timeline.

Under that view, whatever appears earlier tends to be interpreted as cause, and whatever appears later as effect.

But in the framework of this paper, the system does not operate on a single clock.

The fast layer predicts, the slow layer sediments, and the plasticity gate introduces delay and viscosity between the two.

The time differences and report-inaccessibility observed in experiments therefore do not necessarily indicate the absence of consciousness.

They indicate the temporal and pathway conditions required for processing to reach a log-referenceable state.

10.1 Libet-Type Experiments

In Libet-type experiments, subjects are asked to move a wrist or finger at a time of their choosing and to report the moment at which they "decided to move." Brain activity is recorded simultaneously.

As is well known, a readiness potential rises in advance of the movement, and the subject's conscious report of intention appears after it.

Traditionally this result has often been interpreted as: the brain decides first and consciousness learns of it afterward, or alternatively, free will is an illusion.

But this interpretation presupposes that decision, brain activity, and conscious report are placed on the same timeline.

This paper does not adopt that presupposition.

The readiness potential can be read as the formation of an action candidate in the fast layer — a signal that the future-oriented predictive and motor dynamics have begun falling into a certain action attractor.

The report of "I just decided to move," on the other hand, appears after that fast processing has connected to the log-referencing structure of the slow layer.

Only at that point does processing take the reportable form of "I am about to do this."

The time difference observed in Libet-type experiments is therefore not a delay in consciousness but the transformation time between speed scales.

Writing the fast-layer processing speed as v_f and the slow-layer sedimentation speed as v_s:

$$\Delta v = v_f - v_s \tag{45}$$

This speed difference appears as internal delay τ:

$$\tau \propto \frac{\Delta v}{\mu} \tag{46}$$

Here μ is the coupling viscosity between fast and slow layers.

What Libet-type experiments measured was therefore not the presence or absence of free will, but the delay required for fast action preparation to be transformed into a reportable log in the slow layer.

In this view, even the expression "consciousness is delayed" is not quite right.

Consciousness does not operate on the same clock as the fast layer to begin with.

Consciousness opens at the point at which fast processing has been transformed into a log-referenceable state.

For this reason, the earlier appearance of the readiness potential does not show that consciousness is powerless.

It shows that consciousness is not action preparation itself, but the passage point at which action preparation sediments into the slow layer and becomes referenceable.

10.2 Why "Free Will" Is an Ill-Posed Question Here

The reason Libet-type experiments are repeatedly drawn into discussions of free will is the presupposition that "decision" occurs at a single point.

But in the structure of this paper, decision does not localize to a single moment.

In the fast layer, multiple action candidates arise predictively.

The plasticity gate regulates which of them can sediment into the slow layer.

In the slow layer, past logs, meaning gradient, bodily state, and context compose the reportable form.

Compressing this whole into a single "decision point" loses the temporal structure.

It is therefore structurally inappropriate to ask of Libet-type experiments whether "consciousness decided or the brain decided."

The appropriate questions are: which processing preceded in the fast layer; through what delay τ did that processing connect to the slow layer; which logs were referenced by the meaning gradient $\nabla\Phi$; and at what point did a reportable form obtain.

Reframing the questions this way, Libet-type experiments are not experiments that deny free will — they are experiments that make visible the limits of the single-timeline model.

10.3 Split-Brain Experiments

In split-brain experiments, the corpus callosum is severed, restricting information sharing between the left and right hemispheres.

In this situation, appropriate behavioral responses can occur to stimuli presented to the right hemisphere while verbal report of those stimuli may be impossible.

This phenomenon has often been described as evidence for ”two consciousnesses” or ”two selves.”

But in the framework of this paper, this reading is excessive.

What split-brain experiments show is not the doubling of consciousness.

They show that the pathways of processing, sedimentation, and report can become asymmetric.

Even if processing occurs in the right hemisphere, if that processing does not reach the left-hemisphere-dominant verbal reporting pathway, the subject cannot explicitly articulate it.

This does not mean the processing is absent.

The fact that behavioral responses occur shows clearly that local processing is running.

But that processing has not been integrated into the reportable slow log.

What is lost in split-brain experiments is therefore not processing itself, but the referencing pathway.

This point corresponds to Libet-type experiments.

In Libet-type experiments, a temporal delay appears between processing and report.

In split-brain experiments, a pathway disconnection appears between processing and report.

In both cases, the evidence shows that explicit consciousness requires log sedimentation into the slow layer and referenceability.

10.4 Reporting Bottleneck and Log Accessibility

What is particularly important in split-brain experiments is not conflating reportability with consciousness.

Reportability is the pathway by which part of the conscious state is externalized.

But inability to report does not mean processing has not occurred.

In the framework of this paper, explicit report requires at least three conditions:

First, processing has occurred in the fast layer.

Second, that processing has sedimented into the slow layer and become log-referenceable.

Third, that log is connected to the verbal or behavioral reporting pathway.

In split-brain patients, this third condition becomes asymmetric.

As a result, certain processing appears as behavior but not as verbal report.

There is no need to read this asymmetry as ”two consciousnesses.”

More precisely, part of the log-referencing pathway and reporting pathway has been severed.

10.5 Rationalization as Log Completion

A well-known phenomenon in split-brain experiments is that the left hemisphere provides retrospective explanations for actions it could not actually access.

This is often called ”confabulation” or ”rationalization.”

In the position of this paper, this rationalization is not deliberate falsification.

It is the natural behavior of the slow log-referencing system attempting to maintain coherence and fill in missing logs.

The slow layer maintains self-continuity using referenceable logs.

But when the processing pathway is severed, actions occur whose generating logs cannot be referenced.

In this situation, the system generates the most coherent explanation available from referenceable logs rather than leaving the gap as a gap.

This rationalization does not show that consciousness is lying.

Rather, it shows that consciousness constructs coherence from referenceable logs.

In this respect, split-brain experiments weaken the view that the ego is a subject with full access.

The ego is not a center that knows everything.

It is a transformation point that assembles self-continuity from referenceable logs.

10.6 Experimental Correspondence to the Model

Libet-type experiments and split-brain experiments appear on the surface to address different problems.

The former concerns temporal difference; the latter concerns interhemispheric pathway disconnection.

But in the framework of this paper, both are different cross-sections of the same structure.

Libet-type experiments make visible the following relation:

$$\text{fast preparation} \to \text{delayed sedimentation} \to \text{reportable timestamp} \tag{47}$$

Split-brain experiments make visible the following relation:

$$\text{localized processing} \to \text{blocked transmission} \to \text{asymmetric reportability} \tag{48}$$

In both cases, processing itself and conscious report are not identical.

Processing appears as explicit consciousness only after passing through the plasticity gate, delay, sedimentation, and reporting pathway.

Using this structure, the existing experiments are repositioned as follows:

- Libet-type experiments: exposure of speed-scale difference and internal delay τ.
- Split-brain experiments: asymmetrization of log-referencing pathway and reporting pathway.
- Rationalization: slow-layer coherence maintenance in response to inaccessible log gaps.
- Conscious report: externalization of sedimented logs, not processing itself.

Through this repositioning, consciousness is neither the source of causation nor a mere retrospective illusion.

Consciousness is the state that opens when processing has been transformed into a log-referenceable form.

10.7 What These Experiments Do Not Show

Finally, what these experiments do not show should be made explicit.

Libet-type experiments did not show that free will does not exist.

They showed that action preparation and conscious report arise on different time scales.

Split-brain experiments did not show that two selves exist inside a person.

They showed that when processing pathways and reporting pathways are severed, log referencing becomes asymmetric.

Rationalization did not show that consciousness always speaks falsely.

It showed that the slow layer fills in gaps from inaccessible logs in a coherent form.

These experiments therefore do not erase consciousness.

Rather, they return consciousness to its correct position: not the origin of action, not an object generated in the brain, but the state in which log referencing across speed scales is open in a reportable form.

10.8 Summary

This section reinterpreted Libet-type and split-brain experiments from the log-referencing model of this paper.

Common to both is the non-identity of processing and conscious report.

Processing runs. But it does not appear as explicit consciousness unless it sediments into the slow layer, becomes referenceable as a log, and connects to the reporting pathway.

What the existing experiments were therefore showing was not the powerlessness of consciousness.

It was the temporal and pathway conditions required for consciousness to obtain.

11 Qualia as Boundary Phenomenon

This section positions qualia within the structure of this paper.

The aim is not to deny qualia.

Nor is it to place qualia at the center of consciousness.

In the position of this paper, qualia are not consciousness itself.

They are better understood not as objects, but as the phenomenal foreground that appears at the boundary where log referencing is in the process of opening.

11.1 Why the Qualia Problem Fails to Converge Structurally

The qualia problem has long been placed at the center of consciousness research.

The redness of red. The painfulness of pain. The resonance of sound. The sense of being here now.

These are genuinely experienced. They cannot be denied.

But treating qualia as the substance of consciousness makes convergence difficult.

The reason is simple: qualia are not stored objects.

Asking of qualia "where are they?", "in what form are they stored?", "which representation do they correspond to?" is to start the question with a category error.

It is akin to trying to find a flowing boundary as if it were a fixed object.

This paper treats the non-convergence of the qualia problem not as mysteriousness but as a positioning error.

Qualia are not something existing inside.

They are a temporary phenomenon appearing at the boundary where processing is in the process of sedimenting as a log.

11.2 Qualia as Sedimentation Frontier

This paper positions qualia as the sedimentation frontier.

In the fast layer, predictions, reactions, errors, and action candidates run volatilely.

In the slow layer, a portion of these sedimented as logs and become re-referenceable structures.

Qualia do not belong entirely to either layer.

They appear at the boundary where fast-layer processing is in the process of sedimenting into the slow layer.

This boundary can be called the sedimentation frontier, or closure frontier.

"Closure" here does not mean complete closure.

If processing has completely closed, it is already a log, memory, label, or concept.

If it never closes at all, processing flows away without leaving a trace.

Qualia are in between.

Tending toward closure, but not yet completely closed.

Flowing, but already moving toward sedimentation.

This transitional state is what is experienced as qualia.

Figure 6 positions qualia as a transient boundary state.

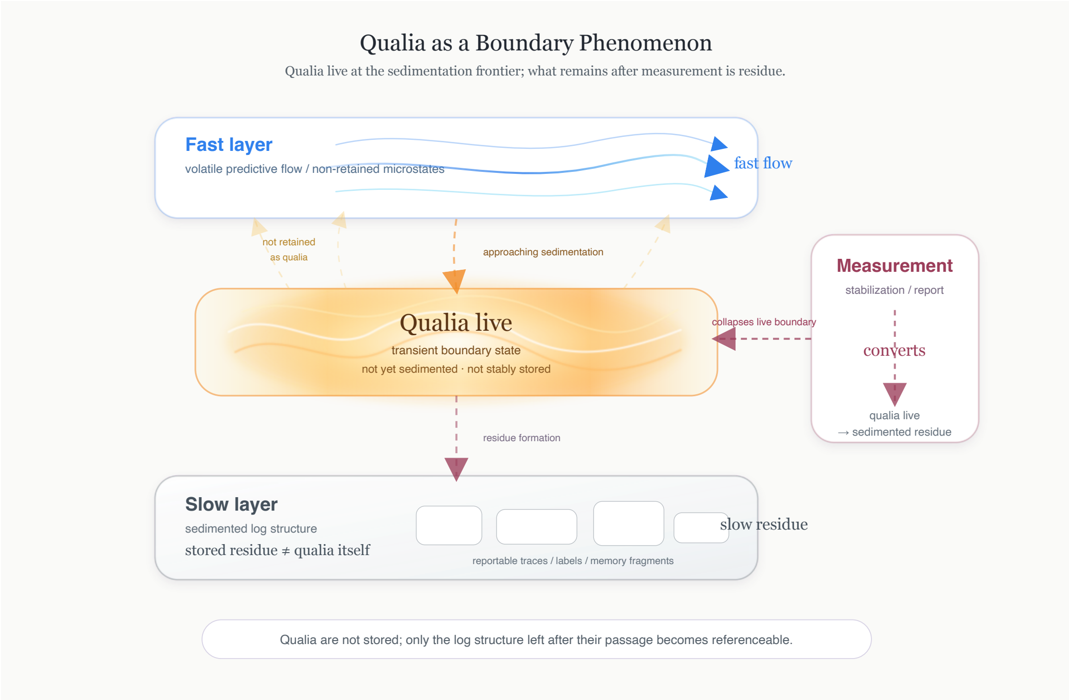

Figure 6: Qualia as a boundary phenomenon. Qualia live at the sedimentation frontier between fast predictive flow and slow log structure. Measurement stabilizes the live boundary and converts it into sedimented residue. What remains referenceable is not the quale itself, but the log structure left after its passage.

11.3 Qualia Are Live, Not Stored

Qualia are not stored.

This may sound strong. But the meaning is not that qualia do not exist.

Qualia exist. They exist, however, not as stored objects but as live boundary states.

The memory of pain persists. The memory of having seen red persists. The memory of having heard music persists.

But what persists is not qualia themselves.

What persists is the log structure that remains after qualia have passed through.

What is replayed as memory is not the former qualia but the slow-layer trace updated by the passage of qualia.

An attempt to faithfully store qualia therefore fails structurally.

The moment of storage, the object is already a log, no longer qualia.

11.4 Qualia and the Plasticity Gate

Qualia appear near the plasticity gate.

For fast-layer processing to sediment into the slow layer, it must pass through the plasticity gate.

If the gate is completely closed, processing leaves no experiential trace.

If the gate is completely open, processing sediments excessively and tends toward fixation.

Qualia appear in the state where the gate is opening, but sedimentation has not yet completed.

Qualia are therefore not products of the gate.

They are the phenomenal surface of processing in the process of passing through the gate.

In this sense, qualia are both a result of log referencing and a mid-point of log sedimentation.

Entirely on the fast-layer side, they are not yet reportable.

Entirely inside the slow layer, they have already become memory or linguistic label.

Qualia appear live only at that boundary.

11.5 Qualia and Effective Torque

The clarity of qualia is not determined by stimulus intensity alone.

In the framework of this paper, qualia vary with the combination of speed difference, coupling viscosity, and meaning gradient:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla \Phi| \tag{49}$$

When $\mathcal{T}_\Phi$ is too small, processing lacks sufficient referencing direction and qualia are diffuse.

When $\mathcal{T}_\Phi$ is in the intermediate range, processing is sedimenting into the slow layer but is still maintained as a live boundary. Here qualia are clear without becoming over-fixed.

When $\mathcal{T}_\Phi$ is too large, processing is drawn excessively into specific meaning wells. Qualia may feel vivid, but their vividness does not necessarily indicate a free conscious state. Rather, an overly strong meaning gradient causes qualia to become fixed to specific memories or narratives.

Clarity of qualia and stability of consciousness are therefore not synonymous.

11.6 Measurement Converts Qualia into Residue

Qualia change character under measurement.

This is not because qualia are mysterious.

It is because qualia are boundary phenomena.

Measurement is the operation of stabilizing an object, making it reportable, and making it comparable.

But qualia exist at the boundary before stabilization.

When measurement is applied, qualia sediment into the slow layer, converting into logs, labels, reports, numbers, and neural summaries.

What remains after measurement is not qualia themselves.

It is the residue left after qualia have passed through:

$$\text{qualia live} \rightarrow \text{measurement} \rightarrow \text{sedimented residue} \tag{50}$$

The claim that qualia cannot be measured therefore does not mean "nothing can be observed."

What can be observed is not qualia themselves but the traces left after qualia have sedimented.

11.7 Why Qualia Feel Central

Qualia are not the substance of consciousness.

Yet experientially they feel very central.

The reason is that qualia appear at the live boundary of log referencing.

In the moments when humans feel "I am conscious right now," some processing is in the process of sedimenting, while at the same time it has not yet fully become a log.

At that boundary, sensation appears vividly.

Qualia therefore appear to be the cause.

But structurally, qualia are not the cause.

They are the foreground observed at the boundary between processing and sedimentation.

As light appears not as the object itself but as the relation between the object and observational conditions, qualia appear as the relation between processing and log sedimentation.

Placing qualia at the center mystifies consciousness.

But placing qualia at the boundary allows consciousness to be re-read as a log-referencing structure.

11.8 Relation to the Hard Problem

The so-called hard problem arises from treating qualia as fixed objects of explanation.

Why does this neural activity accompany "this feeling"? Why do physical processes become subjective experience?

These are powerful questions.

But in the position of this paper, they treat qualia as stored objects.

This paper treats qualia not as stored objects but as the live foreground at the sedimentation boundary.

In that case, what should be asked changes.

Not "where are qualia?" but "under what conditions is processing experienced as a boundary just before being logged?"

This question can be answered through the structure of speed difference, coupling viscosity, internal delay, meaning gradient, and plasticity gate.

This paper therefore does not claim to have solved the hard problem.

Rather, it moves the hard problem from an unsolvable problem-of-objects to a describable problem of boundary conditions.

11.9 Summary

This section positioned qualia as a boundary phenomenon.

- Qualia are not the substance of consciousness.
- Qualia are not stored objects.
- Qualia appear at the sedimentation frontier between fast and slow layers.
- Qualia occur as a live phenomenon near the plasticity gate.
- Measurement converts qualia into residue.
- The clarity of qualia depends on Δv, μ, τ, $\nabla\Phi$, $\mathcal{T}_\Phi$.
- The hard problem does not converge as long as qualia are sought as objects.

Qualia are therefore the live, non-storable foreground that appears at the boundary where log referencing is in the process of sedimenting.

12 Opening Model

This section organizes the central thesis of this paper — consciousness is not modeled as a generated object — as an opening model.

"Opening" here is not a metaphorical expression.

It is a structural description: when the fast layer, slow layer, plasticity gate, meaning gradient, and internal delay enter a certain relation, a log-referenceable passage is temporarily opened.

Consciousness is not a new object created inside.

Consciousness is the state in which referenceability has opened between already-running processing and already-sedimented logs.

12.1 The Generation Assumption

Most discussions of consciousness adopt a generation model, explicitly or implicitly.

In the generation model, consciousness is treated as something produced by some processing.

Under this view, the following questions naturally arise: which brain region generates consciousness; which circuit produces it; which computational quantity gives rise to it; how much consciousness is generated; can generated consciousness be stored or copied?

These questions seem natural at first glance.

But they all contain the presupposition of treating consciousness as a product: a generator, a generation process, and a generated object.

This three-term structure is the foundation of the generation model.

This paper does not adopt that presupposition.

Consciousness is not treated here as the product of processing; it is the state that opens when the relation between processing and logs enters a certain temporal condition.

12.2 From Product to Relation

In the generation model, consciousness is treated as a product.

In this paper, consciousness is treated as a relation.

That is, consciousness is the referencing relation established between fast predictive processing and slow log sedimentation.

For this relation to obtain, at least the following conditions are required:

1. Processing is running in the fast layer.
2. Referenceable logs have sedimented in the slow layer.
3. There is a finite speed difference Δv between the two.
4. Coupling viscosity μ lies between complete synchrony and complete separation.
5. Internal delay τ is maintained as referenceable thickness.
6. Meaning gradient $\nabla\Phi$ is providing a referencing direction.
7. Effective torque-like quantity $\mathcal{T}_\Phi$ lies within the consciousness window.

When these conditions are met, consciousness is not ”produced.”

Log referencing opens.

Consciousness is therefore not an output but an opening of a relation.

12.3 Opening Through the Plasticity Gate

The plasticity gate plays a central role in the opening model.

The plasticity gate is not a device that produces consciousness.

It is the boundary condition regulating whether fast-layer processing can sediment into the slow layer, and to what degree slow-layer logs influence current processing.

If the gate is completely closed, processing does not reach the slow layer. Even though processing runs, log referencing does not open.

If the gate is completely open, processing sediments excessively and tends to lose plasticity. Consciousness tends less to open freely and more to become fixed to specific logs or meaning wells.

What the opening model requires is not complete opening.

Rather, it is partial and temporary opening.

This opening is regulated by speed difference, coupling viscosity, internal delay, and meaning gradient:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla\Phi| \tag{51}$$

When $\mathcal{T}_\Phi$ lies within the consciousness window, the gate opens in a form appropriate for log referencing.

At that point, the conscious state obtains.

12.4 Consciousness Opens, Qualia Appear

As seen in the previous section, qualia are the live foreground appearing at the boundary where log referencing is in the process of sedimenting.

In the opening model, qualia too are not products.

Qualia are the phenomenal surface that appears at the boundary when consciousness opens.

For consciousness to open is for a referenceable passage to be established between the fast and slow layers.

At the boundary of that passage, processing has not yet completely become a log.

But it has not completely flowed away either.

This transitional state appears as qualia.

Qualia are therefore not the cause that produces consciousness.

Qualia are the live foreground observed at the boundary of the consciousness opening.

In the generation model, qualia invite the question "where are they produced?"

In the opening model, the question changes to: under what conditions does the boundary of log referencing appear as a live foreground?

This question can be described through speed difference, viscosity, delay, meaning gradient, and plasticity gate.

12.5 Why Consciousness Cannot Be Stored

From the perspective of the opening model, the idea of storing consciousness is structurally misplaced.

What is stored is not consciousness itself.

What is stored is the log structure updated while consciousness was open.

Equally, what is stored is not qualia themselves.

What is stored is the trace sedimented after qualia have passed through.

Consciousness obtains only while the state is open.

When the opening closes, the conscious state does not persist.

But its traces can remain in the slow layer.

This is why people can remember past experiences.

But what is remembered is not past consciousness itself.

It is the log that was transformed while past consciousness was open.

In this respect, the opening model distinguishes memory from consciousness.

Memory is stored. Consciousness is not stored.

Consciousness opens when stored logs are again referenced by current processing.

12.6 Why Consciousness Cannot Be Copied

For the same reason, consciousness cannot straightforwardly be copied.

What can be copied is structure, records, weights, logs, reports, representations.

But consciousness is not those objects themselves.

Consciousness is the state that opens when those objects enter a specific temporal relation.

Replicating the log structure at a given moment therefore does not guarantee that the replica has the same conscious state.

What is required is not merely log replication.

It is that fast processing, slow sedimentation, plasticity gate, speed difference, delay, meaning gradient, and non-closure loop operate in the same way.

Copying consciousness is therefore not data copying but reproduction of opening conditions.

In this respect, the opening model does not reduce consciousness to information content.

Consciousness is not content but the dynamic establishment of a relation.

12.7 Opening and Non-Closure

The opening model naturally corresponds to VED's non-closure principle.

In a completely closed structure, no opening exists.

In a completely open structure, sedimentation does not obtain.

What consciousness requires is the middle.

Tending toward closure but not closing completely. Open but not flowing away. Fixed but not over-fixed.

Within this range of temporal non-closure, log referencing opens.

The opening model therefore does not treat consciousness as an incompletely generated product.

It treats consciousness as the opening that obtains when non-closure has stabilized.

In this sense, rather than "consciousness is not generated," more precisely: "consciousness opens through not closing completely."

12.8 Opening and Self

The ego and self are also repositioned within this opening model.

The self is not the subject that owns consciousness.

The self is the referencing pattern that stabilizes as a result of log referencing being repeated and a certain terrain forming in the slow layer.

If consciousness is a temporary opening, the self is the terrain formed after that opening has repeatedly passed through.

The self is therefore not the cause of consciousness.

Rather, self-likeness sediments as consciousness repeatedly opens, logs are updated each time, and meaning gradients form.

In this view, the self is neither illusion nor absolute subject.

The self is the stable structure of the slow layer formed through the repetition of log referencing.

12.9 Summary

This section organized consciousness as an opening model.

- Consciousness is not a product.
- Consciousness is the state in which referenceability has opened between fast processing and slow logs.
- The plasticity gate is not a device that produces consciousness but a boundary condition regulating log-referencing opening.
- Qualia are the live foreground appearing at the opening boundary.
- What is stored is not consciousness but the logs updated while consciousness was open.
- What can be copied is logs and structure, not consciousness itself.
- Consciousness opens within the range of temporal non-closure, not at complete closure or complete opening.
- The self is not the cause of consciousness but the slow-layer terrain formed through the repetition of opening.

The central thesis of this paper can therefore be restated as:

Consciousness is not modeled here as a generated object.

Consciousness obtains when log referencing opens within temporal non-closure.

13 Measurement and Collapse

This section organizes what measurement does to the conscious state, and in particular to qualia.

”Collapse” here does not directly mean the collapse of the wave function in quantum mechanics.

Collapse in this paper refers to the conversion of a live boundary state into a reportable, storable log.

That is, measurement is not an operation that finds consciousness, but an operation that transforms the consciousness opening state and sediments it into the slow layer.

13.1 Measurement Requires Stabilization

Measurement requires stabilization.

To measure something, it must be comparable, repeatable, and recordable.

Whether measuring neural activity, behavior, or subjective report, the object is stabilized in some form.

But the conscious state treated in this paper is not a completely stabilized object.

Consciousness is a temporal non-closure regime that opens when speed difference, coupling viscosity, internal delay, and meaning gradient enter a certain range.

This regime is not a static object but a dynamic relation.

Measurement is therefore not neutral observation.

Measurement is the operation of converting a moving relation into a recordable form.

13.2 Measurement as Plasticity Closure

The most important operation measurement performs is the closing of the plasticity gate.

Qualia appear at the boundary where fast-layer processing is sedimenting into the slow layer.

This boundary has not yet completely become a log.

But measurement converts that boundary into a reportable form.

Reporting. Choosing from options. Quantifying intensity. Recording as brain activity. Describing in language.

All of these are operations that sediment the live boundary state into the slow layer.

Measurement therefore does not extract qualia as they are.

Measurement logs qualia.

At that point, qualia disappear as qualia and a residue remains instead.

13.3 From Qualia Live to Sedimented Residue

This conversion can be expressed as:

$$\text{qualia live} \to \text{measurement} \to \text{sedimented residue} \tag{52}$$

"Qualia live" is the live foreground appearing at the sedimentation frontier.

"Sedimented residue" is the log, report, label, number, or neural summary that remains after the boundary state has been sedimented by measurement into the slow layer.

The crucial point is that the residue is not meaningless.

The logs that remain after measurement can indeed be analyzed, compared, and subjected to statistical treatment.

But they are not qualia themselves.

They are the traces left after qualia have passed through.

What measurement can access is therefore not the substance of qualia but the passage traces of qualia.

13.4 Why Precision Makes Qualia Disappear

Increasing measurement precision can make qualia seem to disappear.

This is not because measurement is imperfect.

Rather, it is because measurement is succeeding.

Precise measurement stabilizes the object more strongly.

The stronger the stabilization, the more readily the live boundary sediments into the slow layer.

As a result, what measurement yields becomes a more definite log, a more definite label, a more definite number.

But that definiteness is not the definiteness of qualia themselves.

It is the definiteness of the residue after qualia have sedimented.

For this reason, measurement clarifies qualia while simultaneously erasing them as qualia.

Here lies the paradox of qualia measurement.

13.5 Measurement and the Consciousness Window

This measurement-induced transformation can also be understood from the perspective of the consciousness window.

A stable conscious state exists in the following intermediate region:

$$0 < |\Delta v| < \Delta v_{\max}, \quad \mu_{\min} < \mu < \mu_{\max}, \quad \tau_{\min} < \tau < \tau_{\max}, \quad \mathfrak{T}_{\min} < \mathfrak{T}_{\Phi} < \mathfrak{T}_{\max} \tag{53}$$

Measurement moves the live state within this window toward a slower, more stable log.

That is, measurement changes the relation among Δv, τ, μ, and $\mathfrak{T}_{\Phi}$.

In particular, report and recording push the boundary state toward the slow layer.

As a result, the measurement target is converted from a live opening into a sedimented residue.

In this sense, measurement is not an operation that peers into consciousness from outside without disturbing it.

Measurement is an operation that changes the regime of the conscious state.

13.6 Measurement Is Not Error

One point of caution here.

The fact that measurement converts qualia into residue does not render measurement meaningless.

Measurement is not failure.

Measurement is the necessary transformation for handling live boundary states.

The problem is mistaking measurement results for qualia themselves.

Subjective reports, reaction times, neural activity, and linguistic descriptions obtained through measurement are all important.

But they are not boundary phenomena themselves; they are logs after boundary phenomena have sedimented.

The position of this paper is therefore not anti-measurement.

Rather, it is the position of precisely repositioning what measurement is measuring.

Measurement does not measure qualia.

Measurement measures the sedimented structure remaining after qualia have passed through.

13.7 Introspection as Internal Measurement

This structure applies not only to external measurement but also to introspection.

Introspection is the operation of observing, verbalizing, and grasping one's own conscious state.

But introspection is also measurement.

Directing attention to a sensation. Putting it into words. Evaluating its intensity. Considering its cause.

All of these operations sediment the live boundary state into the slow layer.

What introspection captures is therefore also not qualia themselves but the structure remaining after qualia have sedimented.

This does not mean introspection is unimportant.

Introspection is a powerful method of log updating in the slow layer.

But introspection does not store qualia; it logs them.

Confusing this point leads to the illusion that introspection directly sees the substance of consciousness.

13.8 Experimental Implication

In the framework of this paper, experimental design in consciousness research also takes on a slightly different perspective.

What should be measured is not qualia themselves.

What should be measured is under what conditions qualia appear, under what conditions they sediment, and what residue they leave.

The experimentally important quantities are therefore:

1. The delay until live experience is converted into a reportable log.
2. The transformation of experiential content by the strength of report demands.
3. Changes in residue due to differences in semantic load.
4. Changes in τ or $\mathfrak{T}_\Phi$ due to attention manipulation.
5. Changes in neural activity patterns before and after report.
6. How experience is stabilized through verbalization.

These are not attempts to capture qualia directly.

They are attempts to investigate the process by which qualia sediment.

13.9 Relation to the Measurement Problem

The collapse described in this section is not a direct interpretation of the quantum measurement problem.

But there is a structural similarity.

In both cases, measurement is not a neutral operation that preserves the object as it is.

Measurement is the operation of stabilizing a state and converting it into a recordable form.

In the case of qualia, this operation converts the live boundary state into sedimented residue.

The measurement problem in consciousness research is therefore not the problem of where qualia are hiding.

It is the problem of how boundary states are logged by measurement.

13.10 Summary

This section organized measurement not as external observation of the conscious state but as transformation of the opening state.

- Qualia are the live boundary state appearing at the sedimentation frontier.
- Measurement sediments that boundary state into the slow layer.
- What remains after measurement is sedimented residue, not qualia themselves.
- As measurement precision increases, residue becomes more definite, but live qualia are closed.
- Introspection is also internal measurement and logs qualia.
- Measurement is not error but transformation for handling boundary states.
- What should be experimentally treated is not qualia themselves but the conditions and residue of qualia sedimentation.

Measurement is therefore not the capture of consciousness as it is.

Measurement is the operation that closes the temporal non-closure opening and converts the live boundary into slow-layer logs.

14 Non-Localization

This section organizes why consciousness cannot be localized to a specific site.

Non-localization here does not mean consciousness has no neural substrate.

Conscious states clearly depend on bodily and brain activity.

But that dependency does not take the form of consciousness ”existing” at a single site.

In the position of this paper, consciousness is not a location but a temporal regime that opens when speed difference, coupling viscosity, internal delay, meaning gradient, and log referencing enter a certain relation.

An attempt to localize consciousness is therefore likely searching not for consciousness itself but for the residue that remains after consciousness has passed through, or the local substrate that participates in consciousness.

14.1 The Mistaken Search for a Seat

Consciousness research has a recurring tendency to search for a ”seat of consciousness.”

Which brain region generates consciousness? Which circuit produces subjective experience? Which signal corresponds to consciousness itself?

These questions seem natural at first glance.

But behind them lies the generation model: consciousness is produced somewhere, and what is produced exists somewhere.

This paper does not adopt that presupposition.

Consciousness is not a product but an opening state.

It is the state in which referenceability has opened between processing and logs, and its establishment involves multiple time scales, plasticity gate, meaning gradient, and non-closure loop.

The question of placing consciousness at a single point therefore has the wrong object from the start.

14.2 Local Substrates and Non-Local Regimes

However, the non-localization of consciousness does not mean consciousness has no local substrate.

Fast-layer processing has corresponding neural activity.

Slow-layer log sedimentation also has corresponding neural and bodily substrate.

Reporting pathways, memory pathways, sensory processing, motor preparation, and verbalization each have their own local substrates.

These are important.

But none of them individually is consciousness itself.

Consciousness is the regime that opens when local substrates temporally couple and enter a log-referenceable relation.

What is required in consciousness research is therefore not the denial of local substrates.

What is required is not conflating local substrates with the non-local regime.

Brain regions are involved.

But the conscious state obtains not as regions but as the relation among regions, temporal layers, and logs.

14.3 Why Localization Converts Process into Residue

Localization involves the operation of fixing an object.

To assign a phenomenon to a specific location, the phenomenon must be treated as a stable object.

But consciousness is a live opening state, not a completely fixed object.

As seen in §13, measurement does not preserve qualia or the conscious state as they are; it sediments them into the slow layer and converts them into residue.

Localization is the same.

What localization yields is not live consciousness itself but the residue, trace, correlation, or reportable log that was involved in the conscious state.

The presence of activity in a certain region is therefore important.

But it does not mean consciousness exists in that region.

It shows that the region is participating as part of the log-referencing loop or its residue.

Localization is not meaningless.

But what it captures is not consciousness itself but the local conditions that were involved when consciousness opened.

14.4 Consciousness as Temporal Relation

In the framework of this paper, consciousness is not a spatial location but a temporal relation.

The fast layer anticipates the future. The slow layer retains the past. The plasticity gate regulates what sediments and what flows away between them. The meaning gradient directs which logs are easily referenced.

When this relation enters a certain range, the conscious state opens.

The appropriate form of the question about consciousness is therefore not ”where is it?” but ”under what temporal conditions does processing become log-referenceable?”

This condition was expressed as the consciousness window:

$$0 < |\Delta v| < \Delta v_{\max}, \quad \mu_{\min} < \mu < \mu_{\max}, \quad \tau_{\min} < \tau < \tau_{\max}, \quad \mathfrak{T}_{\min} < \mathfrak{T}_{\Phi} < \mathfrak{T}_{\max} \tag{54}$$

This window is not a point in space.

It is a dynamic region in which multiple processing layers and log structures are temporally coupled.

14.5 Neural Correlates Are Not Consciousness Itself

Searching for the neural correlates of consciousness is important.

But neural correlates are not consciousness itself.

Neural correlates indicate the conditions under which conscious states open, or the traces left after conscious states have opened.

For example, if a certain neural activity corresponds to a particular conscious report, that correspondence is important information.

But this alone does not allow saying that neural activity is consciousness itself.

The reason is that the same activity can carry different meaning depending on reporting conditions, attentional state, meaning gradient, delay, and plasticity gate state.

This paper does not deny neural correlates.

Rather, it reads neural correlates as part of the log-referencing loop.

Neural activity is not the sole seat that generates consciousness but one element of the temporal non-closure structure through which consciousness opens.

14.6 Distributed Does Not Mean Diffuse

Saying that consciousness is non-local can be misunderstood as implying that consciousness is spread vaguely throughout the brain.

But the position of this paper is not such a diffusion model.

Non-local does not mean locationless; it means not reducible to a single location.

The conscious state obtains when multiple local processes enter a specific temporal relation.

Consciousness is therefore not everywhere.

Rather, it opens only under quite strict conditions.

Speed difference is required. Viscosity is required. Delay is required. Meaning gradient is required. Plasticity gate is required. And these must reach neither complete closure nor complete separation.

In this sense, consciousness is not diffuse but conditioned.

14.7 From Localization to Conditions

There is no need to abandon the localization question entirely.

But it is not the final form of the question.

The question ”which region is active?” should be transformed into ”which local substrates are participating in the log-referencing loop?”

This question is then extended as follows: which time scales are involved; which speed differences are maintained; how much delay is there; which logs have become referenceable; which meaning gradients are directing referencing; which plasticity gates are opening and closing.

Changing the questions in this way transforms consciousness from a ”location-finding” problem into a condition-arrangement problem.

This is the shift in questioning that this paper proposes.

14.8 Summary

This section organized the non-localization of consciousness.

- Consciousness has neural substrates.
- But the conscious state itself does not localize to a single site.

- What localization captures is not live consciousness but local substrates or sedimented residue.
- Consciousness is the temporal relation formed by speed difference, coupling viscosity, internal delay, meaning gradient, and plasticity gate.
- Neural correlates are important, but they should be interpreted not as consciousness itself but as part of the log-referencing loop.
- Non-local does not mean diffuse; it means not reducible to a single site.
- The appropriate question is not "where is consciousness?" but "under what conditions does the conscious state open?"

The non-localization in this paper is therefore not mystification.

It is the consequence of treating consciousness not as a spatial object but as a temporal non-closure regime.

15 Qualitative Predictions

The framework of this paper is not a theory that measures consciousness as a single quantity.

The predictions presented here are therefore not numerical predictions of a "consciousness quantity" but qualitative predictions about how the conscious state deforms when speed difference, coupling viscosity, internal delay, meaning gradient, and effective torque-like quantity change.

The central relation is the effective quantity introduced in preceding sections:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla \Phi| \tag{55}$$

$\mathcal{T}_\Phi$ is the effective torque-like quantity coupling time-scale difference and meaning gradient. It does not generate consciousness; it is a structural index showing how strongly and in which direction log referencing tends to open.

15.1 Synchronization Collapse

This prediction concerns the case where speed difference is too small:

$$\Delta v \to 0 \tag{56}$$

The boundary between the fast and slow layers collapses into synchrony. Processing stabilizes, but the temporal thickness required for referencing is lost. Conscious access weakens; action tends toward habituation, automation, and reflexive response.

In this state, processing occurs, but the sense of "currently referencing" diminishes.

Excessive synchrony therefore does not strengthen consciousness; it works toward making conscious referencing unnecessary.

15.2 Dissociation by Excessive Speed Differential

This prediction concerns the case where speed difference is too large:

$$|\Delta v| \gg 0 \tag{57}$$

The correspondence between fast and slow layers becomes difficult to maintain. Predictions and associations run in the fast layer, but they do not stably sediment as logs in the slow layer. Conscious access fragments; structures resembling dreaming, daydreaming, dissociation, and hallucinatory states become more likely to appear.

In this state, even though processing intensity is high, the coherence of log referencing decreases. Referencing does not strengthen — it scatters.

15.3 Coupling Viscosity and State Transition

This prediction concerns changes in coupling viscosity μ.

When μ is too low, speed difference does not slip sufficiently and the layers tend to adhere:

$$\mu \to 0 \tag{58}$$

Processing becomes rigid and tends to fix to specific logs or meaning wells.

When μ is too high, the layers slip excessively:

$$\mu \to \infty \tag{59}$$

Fast-layer changes do not transfer sufficiently to the slow layer; log referencing becomes unstable.

The conscious state is therefore most stable in the intermediate range of μ:

$$\mu_{\min} < \mu < \mu_{\max} \tag{60}$$

This prediction suggests the possibility of treating conscious states as a phase diagram: consciousness is not binary ON/OFF but a regime that transitions continuously with coupling viscosity and speed difference.

15.4 Delay Window

This prediction concerns internal delay τ:

$$\tau \propto \frac{\Delta v}{\mu} \tag{61}$$

When τ is too small, processing approaches immediate reaction and disappears before sedimenting into the slow layer ($\tau < \tau_{\min}$). Conscious referencing is weak and action approaches automatic processing.

When τ is too large, the correspondence between current processing and log sedimentation breaks down ($\tau > \tau_{\max}$). Consciousness lags, fragments, and instabilities in self-sense and time sense become more likely.

A stable delay window is therefore required:

$$\tau_{\min} < \tau < \tau_{\max} \tag{62}$$

15.5 Meaning Gradient Bias

This prediction concerns meaning gradient $\nabla\Phi$.

Log referencing is not opened equally for all logs. In regions where the gradient of the information potential Φ is large, processing is easily drawn in that direction:

$$|\nabla\Phi| \uparrow \quad \Rightarrow \quad \text{referencing bias} \uparrow \tag{63}$$

Memories, emotions, concepts, and narratives with strong meaning gradients more easily enter conscious referencing.

But when the meaning gradient is too strong, referencing becomes fixed to specific wells. This can appear as repetitive thought, compulsive attention, fixation, and traumatic re-referencing.

The meaning gradient directs consciousness, but an excessive gradient takes away the degrees of freedom of referencing.

15.6 Qualia and Measurement

This prediction concerns qualia and measurement.

If qualia are boundary phenomena between fast and slow layers, measurement changes that boundary.

Measurement converts live boundary states into slow, reportable logs:

$$\text{measurement} \quad \Rightarrow \quad \text{boundary closure} \quad \Rightarrow \quad \text{sedimented residue} \tag{64}$$

This prediction is not the simple claim that qualia cannot be measured.

It is the claim that what can be measured is not qualia themselves but the log structure remaining after qualia have passed through.

15.7 Measurement-Induced Residue Shift

The stronger the measurement demand, the more live qualia sediment into the slow layer and stabilize as reportable residue.

Increased measurement precision therefore appears not as capture of qualia themselves but as clarification of sedimented residue.

This prediction requires treating measurement not as mere observation but as a regime-changing operation on the conscious state.

Even for the same experience, if the strength of reporting demand, options, verbalization, or numerical evaluation changes, the remaining log structure changes.

Qualia reports obtained in experiments must therefore be interpreted not as live boundary states themselves but as residue shaped by measurement conditions.

15.8 Split-Brain Asymmetry

This prediction concerns split-brain experiments.

Consciousness does not double in split-brain patients.

What is predicted is the asymmetrization of log-referencing pathways.

Even if processing occurs in one pathway, if it cannot reach a slow reportable log, it does not enter explicit consciousness.

The rationalization observed in split-brain patients is therefore not another self speaking but the slow layer supplying coherence for inaccessible log gaps.

15.9 Non-Local Regime Prediction

If conscious states are temporal regimes rather than single sites, then even the same local activity will produce different conscious reports when speed difference, delay, meaning gradient, and reporting conditions change.

Neural correlates cannot alone determine conscious states and must be interpreted together with temporal coupling conditions.

This prediction does not deny local correspondence of neural activity.

Rather, it claims that the same site activity carries different conscious meaning depending on which temporal coupling it participates in within the consciousness window.

Therefore, even with identical site activity, if slow-log sedimentation, connection to reporting pathways, and meaning gradient strength differ, subjective report and behavioral response should differ.

15.10 Consciousness Window as Phase Diagram

The predictions above show that conscious states exist not as a point but as a window.

$$0 < |\Delta v| < \Delta v_{\max}, \quad \mu_{\min} < \mu < \mu_{\max}, \quad \tau_{\min} < \tau < \tau_{\max}, \quad \mathfrak{I}_{\min} < \mathfrak{I}_{\Phi} < \mathfrak{I}_{\max} \tag{65}$$

Within this consciousness window, processing does not fully synchronize or fully separate, and log referencing opens stably.

This window can be read as a phase diagram. Placing speed difference $|\Delta v|$ on the horizontal axis and coupling viscosity μ on the vertical axis, synchronization collapse, dissociation, fixation, fragmentation, and stable log referencing are positioned as distinct regimes.

Internal delay τ and meaning gradient $|\nabla\Phi|$ function as control variables that extend the state on this phase diagram in a depth direction.

The consciousness window is therefore not an ON/OFF boundary but a finite stable region formed by multiple control variables.

Outside the window, conscious states collapse in different directions:

- Collapse toward synchrony: automation, habituation, and reflexivity strengthen.
- Collapse toward separation: fragmentation, dream-like states, dissociation, and hallucinatory states strengthen.
- Too-low viscosity: fixation to meaning wells strengthens.
- Too-high viscosity: continuity of referencing is lost.
- Too-weak meaning gradient: referencing loses direction.
- Too-strong meaning gradient: referencing becomes fixed.

15.11 Testable Directions

This paper does not treat consciousness as a single directly measurable quantity.

However, directions for verifying conscious state transitions from changes in peripheral quantities are open.

Among verifiable directions, at minimum:

1. Correspondence between the time-scale difference of neural activity and the stability of subjective report.
2. Changes in τ through attentional manipulation.
3. Increase in referencing bias for semantically loaded stimuli.
4. Transformation of qualia-like reports by strengthened measurement and report demands.
5. Changes in speed difference, viscosity, and delay window in sleep, dreaming, anesthesia, and dissociative states.
6. Asymmetry of log-referencing pathways in linguistically non-reportable processing.
7. Clarification of residue and transformation of live qualia reports through measurement demand intensity.
8. Changes in conscious report due to differences in temporal coupling conditions for identical local activity.
9. Phase-diagram classification of conscious states on axes of speed difference and coupling viscosity.

None of these complete the framework of this paper.

Rather, they are entry points for investigating consciousness not as a product to be found but as a non-closure state across time scales.

15.12 Summary

The qualitative predictions of this paper reduce to one point:

Conscious states stabilize when speed difference, coupling viscosity, internal delay, and meaning gradient enter an intermediate effective region:

$$\text{stable consciousness} \quad \Leftrightarrow \quad (\Delta v, \mu, \tau, \mathcal{T}_\Phi, \nabla\Phi) \in \text{consciousness window} \tag{66}$$

This window is not a fixed location.

It is the dynamic region in which temporal non-closure is maintained without rupture.

Consciousness is therefore not modeled here as a generated object.

Consciousness opens when conditions are met, and closes when conditions break.

16 Pathologies and Limits

This section organizes the extent of validity of the framework in this paper and what lies beyond its scope.

"Pathologies" here does not mean medical diagnostic names.

This paper is not a theory for diagnosing psychiatric, neurological, or consciousness disorders.

What is treated here is structural deviation: the directions in which the log-referencing state can collapse when it departs from the consciousness window introduced in preceding sections.

The purpose of this section is therefore not to explain pathology but to clarify the limit conditions of conscious states.

16.1 Pathology as Regime Deviation

In the framework of this paper, stable conscious states are expressed as a window. Outside this window, consciousness does not merely "weaken." Collapse has direction:

- Direction of vanishing speed difference.
- Direction of excessive speed difference.
- Direction of too-low coupling viscosity.
- Direction of too-high coupling viscosity.
- Direction of too-weak meaning gradient.
- Direction of too-strong meaning gradient.
- Direction where live boundaries sediment excessively through measurement or report.

Each brings a different deformation of referencing state.

The crucial point is not linking these directly to disease names.

What this paper treats is not diagnostic classification but regime transition across time scales.

16.2 Collapse Toward Synchronization

The first deviation is the direction where speed difference becomes too small ($\Delta v \to 0$).

The boundary between fast and slow layers collapses into synchrony. Processing stabilizes and becomes efficient, but temporal thickness for referencing is lost.

In this direction, action automates, reactions become smooth, and explicit conscious referencing weakens.

This is not necessarily a bad state. Skilled movement, habituated action, and much reflexive processing are close to this direction.

But as one moves too far toward synchrony, reinterpretation of situations and re-referencing of logs becomes difficult. Consciousness does not so much stop as become unnecessary: processing runs, but the need for log referencing as opening diminishes.

16.3 Dissociation by Excessive Difference

The second deviation is the direction where speed difference becomes too large ($|\Delta v| \to \Delta v_{\max}$ or beyond).

The correspondence between fast and slow layers becomes unstable. Associations, predictions, images, and reactions run in the fast layer, but their stable sedimentation into the slow layer becomes difficult.

In this direction, the conscious state tends to fragment. Dreams, daydreaming, immersive states, dissociative states, and hallucinatory states can be described as part of this regime.

But care is needed here too. This paper does not equate these states. What they share is only the structure of increased speed difference and decreased stability of log referencing.

This section therefore indicates not diagnostic classification but the direction of referencing-structure breakdown.

16.4 Fixation by Low Viscosity

The third deviation is where coupling viscosity μ is too low ($\mu \to 0$).

Fast and slow layers adhere excessively. Speed difference does not slip sufficiently; processing tends to bind strongly to specific logs or meaning wells.

In this direction, states such as repetitive thought, hyperfocus, fixation, and compulsive referencing tend to arise.

The crucial point is that what is occurring here is not merely "strength of meaning."

The problem is that the coupling of meaning gradient and low viscosity causes processing to fall repeatedly into the same well.

Fixation is thus treated not as a content problem but as a viscosity problem of the referencing pathway.

Conscious content may feel vivid in this case, but vividness does not mean high degrees of freedom. Rather, fixation to meaning wells has strengthened and re-interpretability has decreased.

16.5 Fragmentation by High Viscosity

The fourth deviation is where coupling viscosity μ is too high ($\mu \to \infty$).

Fast and slow layers slip excessively against each other. Even though fast processing runs, it does not sufficiently sediment as a slow log.

Experience becomes difficult to retain; self-continuity and temporal coherence weaken.

In this direction, referencing fragmentation, sedimentation failure into memory, and reporting instability can occur.

This paper does not map this to a specific disease. What is stated here is only the structural tendency predicted when coupling viscosity is too high.

Processing exists, but does not stabilize as log referencing. This differs from collapse toward the synchrony side.

16.6 Weak Meaning Gradient

The fifth deviation is where the meaning gradient is too weak ($|\nabla\Phi| \to 0$).

Log referencing loses direction. Even though processing exists, which logs should be referenced and into which conceptual wells to fall becomes unclear.

In this direction, consciousness becomes diffuse, attention becomes scattered, and the outline of experience weakens.

But weak meaning gradient does not necessarily mean functional decline. Rest, diffuse thinking, free association, and creative wandering may utilize this region.

The problem arises when the meaning gradient is so weak that log referencing cannot form a stable terrain.

A weak meaning gradient can be soft exploration or loss of referencing — the difference is determined by the combination of speed difference, viscosity, delay, and gate state.

16.7 Excessive Meaning Gradient

The sixth deviation is where the meaning gradient is too strong ($|\nabla\Phi| \to \infty$).

Processing is strongly drawn into specific meaning wells. Log referencing becomes vivid, but degrees of freedom decrease.

In this direction, states such as emotionally strong memories, traumatic re-referencing, compulsive repetition, and excessive identification with narratives can be described.

The crucial point is that meaning gradient does not only strengthen consciousness.

Gradient directs referencing, but excessive gradient fixes referencing.

Strong meaning does not therefore necessarily mean deep understanding. An overly strong meaning field can rather take away the plasticity of understanding.

In this respect, $\mathcal{T}_\Phi$ is ambivalent: it is both the condition that opens log referencing and, in excess, the condition that fixes referencing.

16.8 Measurement-Induced Collapse

The seventh deviation is where live boundaries sediment excessively through measurement or report.

As stated in §13, measurement does not preserve qualia themselves.

Measurement sediments live boundary states into the slow layer and converts them into reportable residue.

When measurement demands are too strong, the conscious state tends to be treated not as a live opening but as an excessively stabilized log.

What remains is vivid: verbalized, quantified, made comparable. But that vividness is not live qualia themselves. It is the vividness of residue after qualia have sedimented.

This deviation is important in experimental design and introspection. The more one tries to observe, the more the object is deformed. The more one tries to explain, the more the live boundary is logged.

In the framework of this paper, measurement and introspection are therefore also treated as regime-changing operations that alter the conscious state.

16.9 Mislocalization of Consciousness

The eighth deviation is treating consciousness as identical to local substrates or sedimented residue.

As stated in §14, consciousness has neural substrates. But the conscious state itself does not localize to a single site.

What localization captures is local substrates, neural correlates, reportable logs, and post-measurement residue.

These are important. But identifying them with consciousness itself causes the live log-referencing regime to become invisible.

This error appears not only in discussions seeking the seat of consciousness but also in discussions seeking the self or qualia as specific objects within the brain.

This paper does not deny local substrates.

But local substrates can be necessary conditions for consciousness without being consciousness itself.

Consciousness is the state that opens when substrates enter a specific temporal, semantic, and non-closed relation.

16.10 Limits of the Present Framework

The descriptions so far are intended to organize deviations of conscious states structurally.

But this paper has clear limits.

First, this paper is not a medical diagnostic theory. The regimes described here do not directly correspond to psychiatric or neurological classifications.

Second, the measurable definition of $\mathcal{T}_\Phi$ is not yet sufficiently established. The relation $\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla \Phi|$ is at present a structural expression, not a concrete measurable quantity.

Third, how to map the IFGT definitions of Φ, I, J_I, and the local flow $J_{\log}$ onto neural activity, linguistic report, and subjective report remains incomplete.

Fourth, treating qualia as boundary phenomena does not explain all of subjective experience. It repositions the qualia problem without eliminating subjectivity as a whole.

Fifth, this paper does not provide a unified equation of consciousness. Attempting to fix consciousness in a closed equation loses the temporal non-closure that sustains it.

The theoretical scope of this paper lies therefore not in the complete description of consciousness itself but in the specification of the structural conditions under which consciousness can obtain.

16.11 Why the Limits Matter

These limits are both weaknesses of the paper and safety mechanisms of the theory.

Attempting to measure consciousness as a single variable immediately returns the theory to the generation model.

Treating qualia as stored objects loses their character as boundary phenomena.

Linking pathologies directly to diagnostic names loses the flexibility of structural regimes.

Placing consciousness at local sites makes the live log-referencing loop invisible.

This paper therefore deliberately leaves several points unclosed.

Not closing is not vagueness.

It is because the object this paper treats — consciousness — is itself a temporal structure maintained in non-closure.

If the theory rushes to complete closure, it loses the consciousness that is its object.

Making this constraint explicit is the purpose of this section.

16.12 Summary

Pathologies in this paper are not disease classifications but directions of deviation from the consciousness window.

- $\Delta v \to 0$: synchronization, automation, weakening of conscious referencing.
- Excessive $|\Delta v|$: fragmentation, dream-like states, dissociation-like states, hallucinatory-like states.
- $\mu \to 0$: fixation, repetition, fixing to meaning wells through low viscosity.
- $\mu \to \infty$: sedimentation failure, fragmentation of referencing through high viscosity.
- $|\nabla \Phi| \to 0$: disappearance of meaning gradient, weakening of outline.

- Excessive $|\nabla\Phi|$: over-fixation to meaning wells, decrease of plasticity.
- Excessive measurement demands: conversion of live qualia into sedimented residue.
- Excessive localization: mistaking local substrates or residue for consciousness itself.

None of these determine the presence or absence of consciousness in binary terms.

They are structural classifications showing in which direction the conscious state deforms and under which conditions it leaves the stability window.

17 Relation to the Note

This section organizes the relation between this paper and the companion note series.

This paper is not a replacement for the note series.

The note series is an intuitive reading on consciousness, beginning from experience, metaphor, unease, and everyday sensation.

This paper, by contrast, is the theoretical foundation for fixing the structure behind that.

The role of this paper is therefore not to overwrite the note series with theory but to reposition what was intuitively shown in the note series within the structure of temporal non-closure, speed-scale difference, log referencing, and meaning gradient.

17.1 The Note as Intuitive Layer

In the note series, what readers should first encounter is not equations.

What readers should first encounter is a sense of unease about their own conscious experience.

Why does the body begin moving before one thinks one has decided? Why does a sensation change slightly the moment it is put into words? Why do strong memories and emotions repeatedly return to consciousness? Why does consciousness feel as though it is inside oneself, yet cannot be found when searched for?

The note series handles these questions intuitively.

Theoretical terms can be minimal there.

What matters is that readers feel: "Perhaps consciousness is less something generated and more something that opens."

17.2 The Present Paper as Structural Layer

This paper, on the other hand, structures the intuitions presented in the note series.

In this paper, consciousness has been formalized as follows:

- Consciousness is not a product but a log-referencing state.
- Consciousness is not a single site but a temporal non-closure regime.
- Consciousness depends on speed difference Δv between fast and slow layers.

- Consciousness requires coupling viscosity μ and internal delay τ.
- Log referencing is directed by meaning gradient $\nabla\Phi$.
- Effective torque-like quantity $\mathcal{T}_\Phi$ represents the coupling of speed difference and meaning gradient.
- Qualia are not the substance of consciousness but the live foreground at the sedimentation boundary.
- Measurement converts that live boundary into residue.
- Consciousness is non-local; it opens when multiple local substrates enter a specific temporal relation.

These need not be brought to the foreground in the note series.
But they must be retained as the structure behind the note series.

17.3 What Should Flow Back to the Note

What should flow back from this paper to the note series is not all the equations.
What should flow back is the minimal structure that helps readers understand.
Concretely:

1. The expression that consciousness "opens" rather than "is produced."
2. The intuition that consciousness opens between fast processing and slow memory sedimentation.
3. The organization of the ego not as subject but as transformation point of speed difference.
4. The organization of qualia not as stored objects but as live foreground at the sedimentation boundary.
5. The organization that measurement and introspection do not capture qualia directly but log them.
6. The organization that consciousness obtains not as a single brain point but as a temporal condition.

These are sufficiently intuitive for note series readers while also connecting to the theoretical foundation of this paper.

17.4 What Should Not Flow Back Directly

Conversely, some things should not be brought directly from this paper into the note series.

First, equations should not be brought in excessively. Δv, μ, τ, $\mathcal{T}_\Phi$, $\nabla\Phi$ are important structural variables in this paper, but foregrounding all of them in the note series breaks the flow as a reading.

Second, pathological deviations should not be treated as diagnoses. The pathologies of this paper are directions of deviation from the consciousness window, not psychiatric classifications. Even when treating these in the note series, they must be handled carefully as deformations of conscious states, not medical explanations.

Third, IFGT/VED terminology should not be brought too prominently to the foreground. The role of the note series is not to make readers understand theory names but to let readers experience structure. Theory should support from behind, not advance to the front.

17.5 Layered Translation

The relation between this paper and the note series is not simple summarization.

It is translation across layers.

At L3, consciousness is written:

$$\text{Consciousness} \sim \text{Log Referencing}(\Delta v, \mu, \tau, \mathcal{T}_\Phi, \nabla\Phi) \tag{67}$$

But in the note series, there is no need to present this directly.

In the note series, it can be restated as:

Consciousness is the passage that temporarily opens between fast-flowing processing and slowly sinking memory.

This translation is important. The same structure is shown at different resolutions.

This is the relation between the note series and this paper.

17.6 Relation to Appendices

Between the note series and this paper, supplementary materials and intermediate explanatory layers are placed.

This paper fixes structure as L3.

Supplementary materials explain that structure to readers as L2.

Where necessary, L2.5 serves as a bridge that returns the meaning of equations to intuition.

The note series is the experiential layer readers encounter first, as L1.

Not conflating these layers is important.

L3 must not dominate L1.

L1 must not blur L3.

Each layer has a different role.

The purpose of this paper is not to harden the text of L1 but to fix the structure behind it so that L1 can write with freedom and ease.

17.7 Summary

The relation between this paper and the note series is repositioning, not replacement.

- The note series is the layer of experience and intuition.
- This paper is the layer of structure and equations.
- Not everything in this paper needs to be brought into the note series.
- What should flow back to the note series is the minimal structure that helps readers' intuition.
- Theoretical terms and equations support through supplementary materials and intermediate layers when needed.
- L3 does not overwrite L1.
- L3 supports L1 from behind.

This paper is therefore not the completed version of the note series.

This paper is the skeleton behind the note series — enabling the note series to breathe freely as a reading.

18 Discussion

This section organizes the positioning of this paper, differences from existing theories, theoretical implications, and future tasks.

The central claim of this paper is that consciousness should not be treated as a product.

Consciousness is not modeled here as an object produced somewhere.

Consciousness is a log-referencing state that opens when a certain speed difference, coupling viscosity, internal delay, and meaning gradient obtain between fast predictive processes and slow log-sedimentation processes.

This view does not erase consciousness.

Rather, it removes consciousness from its conventional positions — as source of causation, object within the brain, content of representation, container of qualia — and repositions it as a temporal non-closure regime.

18.1 Relation to Global Workspace-Type Models

In Global Workspace-type theories, consciousness is often understood as information becoming globally available, or being broadcast to multiple processing systems.

This paper can connect with this orientation in part.

In particular, the point that conscious states obtain not at a single site but as a relation among multiple processing systems has affinity with the idea of global availability.

But this paper does not define consciousness as ”broadcasting” or ”sharing” itself.

What this paper emphasizes, before information is widely used, is in which time scale it becomes referenceable.

Fast-layer processing sediments into the slow layer and becomes referenceable as a log.

At that point, the conscious state opens.

Spatial breadth is therefore less important in this paper than temporal transformation.

Information becoming broadly available may be part of the result or conditions of conscious state, but it is not consciousness itself.

18.2 Relation to IIT-Type Models

In IIT-type theories, consciousness is associated with the degree of information integration or structural indivisibility.

This paper also does not reduce consciousness to simple local processing, and emphasizes that multiple elements form a relational structure.

In that sense, treating consciousness as structural relation rather than isolated parts has some common ground.

But this paper does not define consciousness as an integration quantity.

Consciousness is treated not as the magnitude of integrated information but as the temporal condition under which log referencing opens.

If integration is too strong, structure becomes fixed and may lose plasticity.

If integration is too weak, processing dissipates and log referencing cannot obtain.

In this paper, therefore, the intermediate state that does not reach complete closure is emphasized over integration itself.

Consciousness is not maximum integration but a non-closure stability window.

18.3 Relation to Higher-Order and Self-Model Theories

In Higher-Order and self-model theories, consciousness is explained as one representation capturing one's own state, or as a self-model representing a state.

This paper acknowledges the importance of self-reference and re-referencing.

Indeed, for log referencing to obtain, current processing must have a relation to slow-layer logs, and that relation must in turn influence current processing.

In this sense, this paper does not deny self-reference.

But this paper does not hold that self-representation generates consciousness.

The self and ego are stable structures of the slow layer formed through the repetition of log referencing, and are not the cause of consciousness.

The ego is not a subject but the position at which speed difference is transformed.

Self-models are important, but they are terrain formed within the log-referencing loop, not the source of consciousness generation.

18.4 Relation to Predictive Processing and Free Energy Approaches

This paper emphasizes prediction, error, and forward models in the fast layer.

In this respect, it can connect with predictive processing and the free energy principle.

The fast layer anticipates the future, generating action candidates and error corrections.

The slow layer sediments part of this and changes the terrain for subsequent prediction.

This loop contains the structure of prediction and updating.

But the focus of this paper is not the accuracy of prediction or minimization of free energy itself.

What matters is how prediction sediments into a log-referenceable structure.

Even with fast prediction, if it does not reach the slow layer, the conscious state does not open.

Conversely, if sedimentation is too strong, prediction becomes fixed to specific meaning wells.

This paper therefore does not view predictive processing as a sufficient condition for consciousness.

Prediction may be a necessary element, but conscious states require speed difference, viscosity, delay, meaning gradient, and non-closure.

18.5 Why the Model Is Not Reductionist

This paper does not treat consciousness as a mysterious entity.

But this does not mean reducing consciousness to simple physical quantities or neural activity.

Rather, to avoid simple reduction, this paper treats consciousness as a state domain.

Consciousness is not I itself, not J_I, not $J_{\log}$, not Φ, not any one of Δv, μ, τ, $\mathfrak{T}_\Phi$.

Consciousness is the regime that opens when these enter a certain relation.

In this sense, the position of this paper is neither eliminativist nor dualist.

It does not add consciousness as a special entity.

But it does not collapse consciousness into a single physical quantity either.

It treats consciousness as the relation among multiple layers.

18.6 Why the Model Is Not Anti-Neuroscience

This paper holds that consciousness does not localize to a single site.

But this is not a negation of neuroscience.

Rather, it is a framework for interpreting neuroscientific findings more carefully.

Certain brain regions, circuits, signals, oscillations, and activity patterns can certainly correlate with conscious states.

These are important local substrates.

But those local substrates cannot be identified with consciousness itself.

In the position of this paper, neural activity is interpreted as part of the log-referencing loop.

Which activity corresponds to the fast layer? Which activity corresponds to sedimentation into the slow layer? Which pathway supports reportability? Which activity forms the meaning gradient?

Asking in this way repositions neuroscientific data not as searching for the seat of consciousness but as investigating the conditions under which conscious states open.

18.7 The Role of IFGT and VED

In this paper, IFGT and VED have been used not as completed theories that fully explain consciousness but as a coordinate framework for repositioning consciousness.

IFGT provides the structure by which log referencing is directed along the meaning gradient, through projected uses of information density I, information flow J_I, local log-referencing flow $J_{\log}$, and information potential Φ.

VED provides the structure of difference, flow, and persistence in which complete closure is not realized as an ordinary state.

Connecting these two, consciousness is positioned as follows:

Consciousness is modeled as the state in which log referencing directed by the projected information potential is open within a temporal non-closure regime that reaches neither complete synchrony nor complete separation.

This formulation does not reduce consciousness to a physical field.

It is a coordinate system for fixing the variables and relations required when speaking of consciousness.

18.8 Theoretical Implications

The theoretical implications of this paper include several points.

First, the question in consciousness research shifts from "where is consciousness generated?" to "under which conditions does log referencing open?"

Second, qualia are repositioned not as the substance of consciousness but as the live foreground at the sedimentation boundary.

Third, measurement is understood not as capturing consciousness as it is but as an operation converting live openings into sedimented residue.

Fourth, neural correlates of consciousness are interpreted not as consciousness itself but as local conditions participating in the log-referencing loop.

Fifth, pathologies and altered states can be structurally organized not as disease classifications but as directions of deviation from the consciousness window.

These do not negate consciousness research.

Rather, they are translations that move existing data and discussions from the generation model to the opening model.

18.9 Remaining Questions

Unresolved tasks remain in this paper.

First, the measurable definition of $\mathcal{T}_\Phi$ is incomplete. How to map speed difference, coupling viscosity, and meaning gradient onto neural activity, bodily states, subjective reports, and linguistic logs is a future task.

Second, how to map IFGT's I, J_I, Φ and the local flow $J_{\log}$ onto brain data and behavioral data must be further refined.

Third, how to empirically estimate the consciousness window remains undetermined.

Fourth, methodological ingenuity is needed for how to experimentally handle qualia sedimentation residue.

Fifth, how the self, language, and social interaction extend the log-referencing structure exceeds the scope of this paper.

These tasks do not negate this paper.

Rather, they show that this paper not only remains unfinished as a theory but also corresponds to the non-closure of consciousness itself as an object.

18.10 Toward Consciousness, Self, and Intelligence

This paper treated consciousness as the opening of log referencing.

But this framework does not close with consciousness alone.

The self can be understood as the stable terrain formed in the slow layer through the repetition of openings.

Intelligence can be understood as the structure that uses that terrain to extend future prediction, language, social logs, and interaction with the external environment.

In this sense, this paper is not the endpoint of consciousness theory but the bridge to self theory and intelligence theory.

Consciousness is not modeled as a generated object. Consciousness opens.

The self sediments through the repetition of that opening.

Intelligence flows back into the world again using that sedimented terrain.

Following this flow is the next task.

18.11 Summary

This section organized the positioning of this paper.

- This paper treats consciousness not as a product but as the opening of log referencing.
- It has points of contact with Global Workspace, IIT, Higher-Order, and Predictive Processing discussions, but is not identical to them.
- This paper does not reduce consciousness to a single site, single quantity, or single representation.
- It does not deny neuroscience; it reinterprets neural correlates as local conditions of the log-referencing loop.
- IFGT and VED are used not as theories that fully explain consciousness but as coordinate systems for positioning it.
- The claim of this paper is shifting the question in consciousness research from "generation" to "opening conditions."

- Unresolved tasks remain, but this corresponds not only to the incompleteness of the theory but also to the non-closure of consciousness as its object.

The significance of this paper therefore does not lie in explaining consciousness as a closed object.

It lies in repositioning consciousness — as a log-referencing state that opens within temporal non-closure — in a form that allows it to be questioned again.

19 Conclusion

This paper has reformulated consciousness not as an object generated internally but as a log-referencing state that opens between processes at different time scales.

The central claim is simple.

Consciousness is not modeled here as a generated object.

Consciousness opens when fast predictive processes and slow log-sedimentation processes couple within a range that reaches neither complete synchrony nor complete separation.

This opening involves speed difference Δv, coupling viscosity μ, internal delay τ, meaning gradient $\nabla\Phi$, and effective torque-like quantity $\mathcal{T}_\Phi$.

But none of these individually is consciousness itself.

Consciousness is the temporal non-closure regime that obtains when these enter a certain relation.

19.1 What Has Been Proposed

This paper first introduced SSO as the minimal temporal structure for the establishment of consciousness.

SSO is a structure in which processing layers running at different speeds are coordinated without central control.

The fast layer anticipates the future; the slow layer sediments the past.

A finite speed difference remaining between them opens a referencing relation between current processing and past logs.

Next, Torque Converter Attention was introduced.

This treats attention not as selection or weighting but as a temporal mechanism that transforms speed difference without rupture.

In this position, the ego is understood not as a subject but as the position at which the speed difference between fast and slow layers is transformed.

Furthermore, using IFGT and VED, log referencing was repositioned within meaning gradient and non-closure dynamics.

IFGT describes which logs are easily referenced through projected uses of information density I, information flow J_I, local log-referencing flow $J_{\log}$, and information potential Φ.

VED provides the structure of difference and flow in which complete closure is not realized as an ordinary state.

Connecting these two, consciousness was described as the state in which log referencing directed by the meaning gradient is open within a temporal non-closure regime.

19.2 What Has Been Repositioned

This paper repositioned several important concepts.

First, consciousness was repositioned not as an object but as a state domain.

Consciousness is not an object stored somewhere, nor an output generated within the brain.

It is a regime that opens when processing and logs enter a referenceable relation.

Second, qualia were repositioned not as the substance of consciousness but as the live foreground at the sedimentation boundary.

Qualia are not denied.

But they are not stored objects.

What is stored is not qualia themselves but the log structure remaining after qualia have passed through.

Third, measurement was repositioned not as an operation that captures consciousness as it is but as an operation that converts the live boundary into sedimented residue.

Measurement is not meaningless.

But what measurement captures is not consciousness itself but the traces remaining after consciousness opened.

Fourth, non-localization was repositioned not as mystification but as a consequence of treating consciousness as a temporal relation.

Consciousness has neural substrates.

But the conscious state itself does not localize to a single site.

It opens when multiple local substrates enter a specific temporal, semantic, and non-closed relation.

Fifth, pathologies and altered states were repositioned not as disease classifications but as directions of deviation from the consciousness window.

Through this organization, the conscious state is treated not as a binary presence or absence but as a dynamic regime with a direction of collapse.

19.3 The Central Formula

The central structure of this paper can be summarized in the following expression:

$$\text{Consciousness} \sim \text{Log Referencing}(\Delta v, \mu, \tau, \mathfrak{T}_\Phi, \nabla\Phi) \tag{68}$$

This expression does not compute consciousness.

It is the coordinate showing the structural variables required when speaking of consciousness.

More concretely, the conscious state stabilizes within the following window:

$$0 < |\Delta v| < \Delta v_{\max} \tag{69}$$

$$\mu_{\min} < \mu < \mu_{\max} \tag{70}$$

$$\tau_{\min} < \tau < \tau_{\max} \tag{71}$$

$$\mathcal{T}_{\min} < \mathcal{T}_{\Phi} < \mathcal{T}_{\max} \tag{72}$$

Within this consciousness window, processing does not fully synchronize or fully separate, and log referencing opens stably.

Outside the window, conscious states deform in different directions.

Consciousness is therefore not a point but a window.

Not an object but a regime.

Not a product but an opening.

19.4 Why the Model Remains Open

This paper does not provide a complete unified equation of consciousness.

This is not an omission but an intentional constraint.

Attempting to define consciousness as a closed object loses the non-closure that sustains it.

Treating qualia as stored objects loses their character as boundary phenomena.

Identifying neural correlates with consciousness itself makes the structure as temporal regime invisible.

Linking pathologies directly to diagnostic names loses the flexible structure of directions of deviation from the consciousness window.

This paper therefore deliberately does not close completely.

Not closing completely is not abandonment of explanation.

It is because consciousness itself — the topic of this paper — obtains as a temporal structure that does not close completely.

Theory resembles its object.

If consciousness is non-closed, consciousness theory too must not rush to complete closure.

19.5 Toward Further Work

Future tasks are clear.

First, $\mathcal{T}_{\Phi}$ must be brought closer to concretely measurable quantities connected to neural activity, bodily states, subjective reports, and linguistic logs.

Second, methods for mapping IFGT's I, J_I, Φ and the local flow $J_{\log}$ onto brain data and behavioral data must be refined.

Third, the consciousness window must be examined against states such as sleep, dreaming, anesthesia, immersion, dissociation, skilled movement, and verbal report.

Fourth, experimental designs for handling qualia sedimentation residue must be developed.

Fifth, how the self, language, and social interaction extend the log-referencing structure must be addressed.

This last task is the bridge from the consciousness paper to the self paper and intelligence paper.

Consciousness is the opening of log referencing.

The self is the terrain that sediments into the slow layer after that opening has been repeated.

Intelligence is the motion that uses that sedimented terrain to extend into future prediction, language, social logs, and interaction with the external environment.

19.6 Final Statement

Consciousness is not produced somewhere.

Consciousness is the passage temporarily opened between fast-flowing processing and slowly sinking logs.

There is speed difference.

There is delay.

There is viscosity.

There is meaning gradient.

And there is flow that does not close completely.

While this passage is open, processing does not merely flow — it references past logs, updates the terrain of the self, and reconstructs the relation with the world.

This state is what we call consciousness.

The conclusion of this paper therefore returns to the proposition with which it began.

Consciousness is not modeled here as a generated object.

Consciousness obtains when log referencing opens within temporal non-closure.

20 Correspondence Map: This Paper ↔ Conceptual Topography

For readers who will also read Conceptual Topography (CT) alongside this paper, the correspondence between the main concepts of both papers is shown here.

This paper and CT treat different projected cross-sections of the same VED×IFGT framework. Variables are related, but they are not simply identical across layers; neither paper grounds the other. Both describe, from different observation windows, how consciousness as an opening state and concepts as stabilized terrain can be read as projected regimes of the same generative grammar.

This paper	CT (Conceptual Topography)	Shared variables / correspondence
Slow-layer sedimentation structure	Information potential Φ landscape	Both converge as Φ^*, the long-time attractor
SSO (temporal non-closure)	Invisible GPU (landscape operating below consciousness)	Φ constrains thought without being directly referenced by consciousness
TCA (transformation of speed difference)	Token coupling $\lambda \sum_i \ell_i \cdot \nabla\Phi$	Isomorphic structures transforming speed/spatial difference
Log-referencing direction by meaning gradient $\nabla\Phi$	Concept definition: locally stable region $\nabla\Phi \approx 0$	$\nabla\Phi$ simultaneously governs conscious bias and concept formation
Log referencing (state where consciousness opens)	Traversal cost reduction (state where understanding obtains)	Both described as referencing/traversal of $\nabla\Phi$ structure
Qualia (live foreground at sedimentation boundary)	Insight (rapid topological transition)	The same boundary phenomenon observed at different temporal granularities
$I(x, \tau) > 0$ (exclusion of complete closure)	$I(x, \tau) > 0$ (exclusion of complete conceptual closure)	Positivity marks the exclusion of ordinary complete closure in each window; it does not by itself identify consciousness with the conceptual landscape

Reading suggestions:

For readers who read this paper first: reading CT reveals that the ”temporal condition for consciousness to open” in this paper and the ”spatial condition for concepts to form” in CT come from the same structure. What is referenced in a conscious state is determined by the geometry of the conceptual landscape the person holds.

For readers who read CT first: reading this paper reveals that behind CT’s description of ”thought flowing across terrain,” the speed difference between fast and slow layers is required. The terrain is not given; it becomes referenceable only when SSO is established.

Read in mutual reference, both papers suggest that consciousness as an opening state and concepts as stabilized terrain are different projected stabilization regimes of the same VED×IFGT grammar, not identical entities or reducible objects.